UX redesign of the Allodi Accordions Ltd. website

Contents

1. Introduction 5
1.1. Background 5
1.2. Aim of the study 5
1.3. Objective 5
1.4. Research Rationale 6
1.5. Problem Statement 7
1.6. Scope of the study 7
1.7. Significance of the study 8
1.8. Estimated Solution 8
2. Literature Review 9
The Concept of UX 9
Current Usability Heuristics 9
2.6. UX Design Frameworks 11
2.7. Website accessibility – WCAG 14
2.8. Literature summary 15
3. Research Methodology 17
3.1. Research Philosophy 17
3.2. Research Choice 17
3.3. Research Strategy 18
3.4. Research Bias 20
3.5. Time Horizon 20
3.6. Selected UX Design 21
3.7. UX design framework 21
3.8. Issues 23
Chapter 4. Analysis and design 24
4.1. Qualitative Analysis for the target website (http://www.accordions.co.uk) 24
4.2. Evaluation of the target website and its competitors using Nielson's Ten Heuristics 28
Visibility of system status 28
Match between the system and the real world 28
User Control and Freedom 29
Consistency and Standards 30
Error Prevention 31
Minimize memory Load 32
Flexibility and Efficiency of Use 34

Aesthetic and Minimalist Design....34
Help Users Recognize and Recover from Errors....35
Help and Documentation....36
4.3. Evaluation of the target website and its competitors using WCAG Analysis....37
Chapter 5. Implementation and testing....40
5.1. Understanding the user and context of use....40
5.1.1. User Research....40
5.1.2. Persona Development....42
5.1.3 Empathy Building/ User Journey Maps....45
5.2. Specifying user and business requirements....46
5.2.1. System requirements....46
5.2.2. User requirements....46
5.3. Create design solutions....48
5.3.1 User flows....48
5.3.2 Colours, icons, images, language, and tone....48
5.3.3 Wireframing and Prototyping....49
5.4. Testing....63
5.4.1. Usability Testing....63
Chapter 6. Evaluation....69
6.1. Evaluation of the designed system using Nielson's Ten Heuristics....69
Visibility of System Status....69
Match between the system and the real world....71
User Control and Freedom....72
Consistency and Standards....74
Error Prevention....74
Recognition Rather than Recall....75
Flexibility and Efficiency of Use....77
Aesthetic and Minimalist Design....78
Help Users Recognize, Diagnose, and Recover from Errors....79
Help and Documentation....81
6.2. Evaluation of the designed system using WCAG 2.2....81
Perceivable....82
Operable....83
Understandable....85
Robust....88

8. References.....89

1. Introduction

1.1. Background

UX (user experience) and UI (user interface) refer to the inclusive user experience from that specific website, and the interaction between the user and software respectively. Both of these have their major importance in their user experiences that can be either positive, hassle-free or even worse than any other. At present the digital economy largely depends on a user-friendly approach and smooth experiences. This research highlighted the current state of Allodi Accordions Ltd.'s website based on its UX which provides the users in terms of content, design, and usability.

Allodi Accordions Ltd is a company that is registered in Essex Road, Dartford, Kent in London. and it is a private limited company, established on 5 October 2004. The company is an expert in authentic accordions, for commercial use purposes. MacDonald (2013) explained that they commit to maintaining cultural diversity, and traditions along with the standards (Solanki and Pandey, 2019). They serve the users with authentic musical instruments in a digital hub for user interactions. This company is run by a whole family involved but the website has now become a topic to study as it has too many areas to explore. precisely this website is a good option to explore more in detail on website user experience, its flaws also where to improve and make it user-friendly.

1.2. Aim of the study

The study aims to find and fix the existing website that has been built for the company Allodi Accordions Ltd. According to the current upgrades of UI/UX, the company lags with bad experiences and poor visualisations. So, they should improve the intricate parts of user interactions for user satisfaction, more engagement and flexible usability (Blokdyk, 2017). Fixation would be possible by steering user research for classifying pain points, improving the visual appearance, user-centric approach, updating screen presence with modern concepts and mostly implementing user-friendly experiences. The major motive behind all the research is ensuring the clients a user experience that nurtures positive interactions and also exploring the parts to develop so that the platform engagement goes smoothly and effortlessly.

1.3. Objective

- To research and understand available design methods for improving website UX

- To research and evaluate the target website using existing UX design methods
- To identify the UX issues with the target website and strategize to resolve those.
- To design a new prototype of the target website
- To implement the chosen methods for designing the prototype
- To test and evaluate the designed prototype using chosen usability heuristics

1.4. Research Rationale

Background: User Interface UI and User Experience UX of websites play a vital role in client experience. Allodi Accordions Ltd. is a company that has been run by a family for a long time. However, it has a poor visual appeal to the audience. It needs internal changes with the updated time so that the visual presence can draw the attention of customers and convince them to buy products from them (Harris, 2017). This paper is driven by a set of company needs with updated trends. And run for an administration of a landscape within the musical commercial.

Knowledge Gap: From the overall analysis, there is a lot of lacking in comprehensive insights into the website of Allodi Accordions Ltd. Because of its backdated and poor user interface. It has a gap in using the current updated UX guidelines application, where there are too many frameworks, and methods to work with. so, it seems to have a noticeable absence of deep knowledge analysis and application more specifically which is needed for the specific website of the company (Unger and Chandler, 2023).

Goals Overview: This research is conducted to resolve the user interface fixing the internal issues along with targeting to make it more user-friendly so that people can have a seamless interaction with the website and be more convinced to buy products and trust the website. But to make it happen, the goal is to make a bridge between the UX guidelines and the UI of the company. The primary goal here is to make a framework for correcting suboptimal user interface designs, that enhance the UX user experiences, and nurturing a clear understanding of the particulars that come with the most unique user-centric approach

Methods: This research employs a specific research method that works with a qualitative secondary approach, with the use of UX principles and heuristic evolution of existing User interface designs (Hughes and Tarrant, 2019). Throughout the implementations, data taken from different sources this study aims to provide a whole conceptual analysis of the impact and pertinency of Allodi Accordions' user experience UX guidelines which improves the user interface.

Relevance of Research: This research is important for the deep understanding of UX guidelines to enhance the user interface using specific methodologies, and frameworks. This study will contribute to the next relevant research along with offering practical experiments (Urick, Ford and DeMatthews, 2021). For the rising developers, designers and the users mostly. The gaps and lack that the paper finds here through the research will pave the way to solve any problems regarding the user interface so that it can be more effective and user-centric solutions for the company Allodi Accordions Ltd.

1.5. Problem Statement

Although the company has been established and running for many years, they are still struggling with its website and client approach. With time, everything gets updated but this company doesn't meet the current updates of UI/UX which reflect on the bad user experiences of the clients. They indeed have a long history of unique product collections and services but fail to provide captivating and unified user engrossment (O'Neil and Perez, 2013). This dissatisfaction directly hampers the sales growth of the company as clients who are not satisfied don't agree to buy products from such websites because facing trouble circumnavigating the website. Instead, it demotivates people to buy products from them and creates trust issues at the same time.

However, Allodi Accordions Ltd is facing perilous encounters with a suboptimum UI/UX user interface/user experience because of their design which is negatively impacting client interaction, their satisfaction and overall performance. The problem stated here is the necessity of all-inclusive restructuring of the website and attracting the circumnavigating system, functionality. along with ensuring a unified and pleasant involvement for customs interrelating with Allodi Accordions Ltd."

1.6. Scope of the study

The study influences different identified variables. The researcher finds some scope of the study for the paper to fix the current UX of Allodi Accordions Ltd. Such as conducting an evolution of the website and specifying the pain points, where to improve, specifying the sectors of client dissatisfaction thoroughly by an assessment (Forte and Brazdil, 2016). Analysing the market and the competitors and finding their practice of growth, acknowledging and working on them. Along with the evolutions, researching client demand, and current market structures through various types of surveys, and interviews. Most importantly, waged with a set of design principles, and innovative features of Allodi

Accordions Ltd that come with developing different sort of wireframes and interactive prototypes that visualises UX improvements. The study is all about enhancing the outlook of the website and improving the visual design considering colour scales and aesthetics.

Addressing the scope to improve the website, these are some aspects to work on. The research papers offer all-inclusive guidelines to redesign the UX of the company so that it looks more visually appealing, ensuring user centric effective client satisfactory experiences.

1.7. Significance of the study

The study's significance is the key motives behind the restructuring of the existing features of the company. Such as finding the difficulties clients face while entering the website. Most importantly the visual appearance of the website seems to be very messy, analyse the heuristics, an in-depth analysis of the overall user practice. categorising the strengths and weaknesses along with the user impacts. Investigating the existing UX design guidelines along with the current updates. evolute and innovative different website and their user approaches. Designing a high-fidelity prototype, by ensuring proper guidelines for UX design. Estimating the serviceability following Nielsen’s heuristic guidelines made for web submissions. Looking at a sector-specific market player like Allodi Accordions Ltd. may help shed light on the difficulties of digital engagement in general. Digital interactions may vary and depend on each customer's tastes, which might lead to happier and more loyal consumers for businesses that can do this (Rafanell and Sawicka, 2019). From identifying the main problems or concerns that bother the users to gathering their feedback and overall analysis, solving them throughout using the following guidelines of UX. Conducting a gap analysis would compare other existing good websites with good UI can update the website of Allodi Accordions Ltd. Other than that, prioritising the concerns, prototype design will develop the case for the users through testing the usability and testing.

1.8. Estimated Solution

The company must redesign their website for user-friendly interaction.to grab the attention of the targeted audiences it's essential to make the first impression through the website, so it should be attractive, easier to use and more convincing for the clients to buy products from them. A dull boring and difficult website can lose potential customers and fail to draw attention. So, the company should focus on designing the User Interface by following the design concept, and using the methods, frameworks and guidelines.

2. Literature Review

This chapter is about the literature review that is based on the importance of User Experience UX design of websites. This is an overview of so many concepts, ideologies, visualisations, expectations and limitations by intelligent minds, mostly it shows the importance of UX designs. This section includes the study and progression of many hypothetical and impartial data that apply to research findings. It aims to provide a summary of previously attempted data research on the same domain with diverse dimensions, rules, models, concepts, ideologies, guidelines and from different perspectives.

The Concept of UX

User experience UX incorporates all sorts of customer interactions with the organisations, and the product goods or the services they provide. The International Organization for Standardization (ISO) states this experience is the response from the user's viewpoint and perspective and it may vary. UX didn't even exist before 1990. Then after joining Apple, Norman the field of "user experience architect" started growing eventually it extended rapidly with demand to meet the client satisfaction requirements. In the early 90s Norman in an interview with Adaptive Path invented this specific term considering the narrow impact of human interfaces and their usability.

User experiences can be of many types and forms of different elements. They discussed the UX elements that include the crowd of components and the essential concepts that make a website user-friendly. Easier to use and flexible from all aspects for the users. They claimed that user engagement or their interaction with the website largely depends on usability, accessibility and most importantly user-centric approach. It is very difficult for the user to stick with their interest in buying products through a hectic user experience with a lot of challenges. in the research they found the primary objective of UX is making a website flexible, prioritising their comfort and ensuring user enjoyment, which leads to productivity through the testing, evolution, research methods, choices and appropriate applications along with empathy-informed research (Bergstrom and Schall,2014).

Current Usability Heuristics

Nielsen's heuristics

Kraft (2022), said Neilsen's heuristic is founded on Jakob's law. The law draws a line with people who offer most of their time and energy into it. according to the law, it serves as the

foundation for the heuristics. here the users are the main concerns. They desire consistency, reliability and unnoticeable websites and these imitate the established standards. Also, they like to prefer products that are not independent and out of unexpected concerns. However, there are no reliable users to solve such puzzles unless they are experts on it. but the main advantage here is this heuristic is mostly focused on user understanding and longer-term functionality of products. According to Aela (2022), these are some general principles which don't determine Specific rules of usability. The researcher explained the ten heuristics respectively, the system status visibility, the match between platform and reality, user flexibility and controlling, with maintaining consistency with standard.to prevent the errors. Recalling instead of recognizing, diagnosing the errors and recovering them at the same time. Helping users with the documentation. In another research, Gonzalez-Holland et al. (2017) found the interconnection between the development of a website and the accuracy of applying the ten usability heuristics. In this article, the researcher claimed that these are the most widely used tools for designing the interface. As per the study report or outcome, it has been used or has been applied to various modalities of designs (Lyonnais, 2017). This has been modified to suit the environment. However, as this study article was conducted with more than three researchers, they were in between arguing with some conflicting concerns that are about the upgrading with time, as shortly these heuristics may need to change the motive, perspective, and concerns so that it be consistent with the latest updates and solve the problem if it gets any issues with the modern usability.

Shneider man's Eight Golden Rules of Interface Design

Ben Schneider man innovated eight golden rules to design the user interface so that it can be more flexible and easier to use. The rules are respectively striving for consistency, which covers the colour coding scale from the menu hierarchy to standardise information like how it can be conveyed. It plays a vital role in helping users to become familiar with digital platforms and landscapes. And then with the increasing demand for quicker methods to complete the tasks, enabling regular users to use the shortest paths as much as possible is another rule (Wong, 2024). Then it offers informative feedback which solves the complexities while using the websites. Also designing dialogues to harvest the closure, offering minor errors and issues, permitting the easiest action reversals. While using, allowing the users to be the initiator to take actions by supporting the internal locus of control, Reducing the short-term memory load. Following these rules while prototyping or building the design into a reality can make a website far better and more comfortable for the

users. ('Development of research progress tracking system based on Shneiderman’s Eight Golden Rules,' 2023).

Jeff Johnson’s Fourteen Principles of User Interface Design

Baird and Fisher (2005), in their study, stated that, in today's generation, everyone is innovating something ahead of time and updating themselves with the shortest and easiest ways as much as possible. Where this neocolonial generation is large in number with values like being self-paced, and they try to customize the way of learning to be interactive using social networking tools considering the user experience guidelines and the design strategies. These can be possible using or following the principles innovated by Jeff Johnsons. they together formulated the fourteen principles to design the interface. according to the study, this helps enhance user comprehension through the explicit and precise framework Johnson (2013). These principles cover the areas of clarity

2.6. UX Design Frameworks

There are so many UX design frameworks, and they consist of five different elements that are strategy, scope, structure, skeleton and surface. All these elements relate to different individual layers which are involved in creating client experiences. But every layer is dependent on one another.

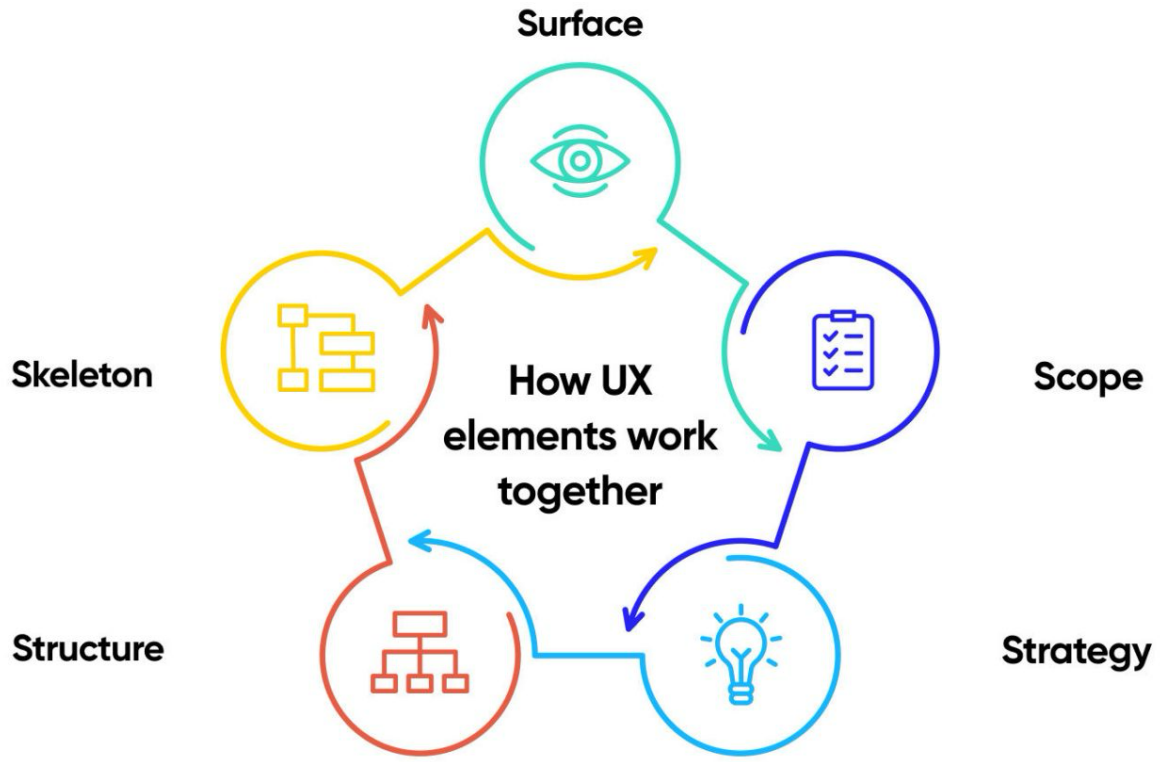


Figure 1: Elements of the User Experience Framework (Podoba, 2024)

Frameworks benefitted in the area of consistency of a website and it assures an identical design. The primary goal of a design framework is to consider the priority of a customer. By integrating client research, compassion, and response, originators or the engineers can create solutions or redesigns that reverberate with clients, recognising their pain points and choices efficiently (Alexiou and Schippers, 2018)

To define user experience, UX Berni and Borgianni (2021) state that, before the 1990s, it did not exist. but according to the demand of user expectations, their necessity, and motivations UX was invented. The interaction between clients and business services artefacts is interpreted in various forms. and UX plays a vital role here in connecting. It appears as a cauldron of relevant and meticulously accompanying ideas. They highlighted the elements of connectivity in all sorts of user experiences (Canziba, 2018). The researchers investigated if it can be built from mapping samples from experiments in design concerns. Also, they classified the frameworks to categorise the experiences of customers

The perception of human-galloped design is nothing new. It has some core similarities with the UX user experiences and the user-centric designs, along with certain differences.

(Dell'Era and Landoni, 2014b). On the other hand, the basic principles of the User interface are internally connected with the user-centric designs. Ballard (2017) stated in his blog article

an inclusive description of the steps that are required, that come with the visual aids, to validate the development of a customised design that fulfils all the unique requirements of each user. and makes it easier to work on, which ensures flexibility. From his perspective, centric designs have more complexity and more restrictions to maintain the overall structure and that is the reason why he said the human-centric design is more comprehensive than UX/UI design. As per Schmidt et al. (2020), Yet again this concept includes a wide array of design elements that include both digital and physical elements. Websites or digital platforms offer designers the necessary tools to develop extraordinary online user experiences. In another article, the researcher found the methods of UCD user-centric design and human-computer interaction HCI together build a positive platform for the users that comes with user experiences in designs of interaction, aligning with the methods of LIDT theories. Which outlines every specific idea to design platforms made for digital learning. The researcher also provides detailed definitions, steps, and stages of all kinds of practices and practical experiments. (Farao *et al.*, 2020)

According to Brown (2024), to accomplish the tasks, setting up a process allows one to complete the tasks quickest way possible and also provides the best results. This model was invented to design a process that was developed in 2005 which is a guideline for understanding the problems that come with it. The researchers found that the framework is divided into four fine stages such as discover, define, develop and how it delivers. But at the beginning of 2004, it was divided into just two major phases which are the problem phase and the solution phase. But in 2019 UK Design Council made some changes to it. Understanding the necessity of the user demand and exploring the problems along with challenges then it helps to identify. In the second step, it finds the solution to the specified problem and allows one to decide and select the important data that is required for solving the problem. In the development and delivery phases, the website gets ready with the solved areas and is delivered to users (Banbury *et al.*, 2021).

Hooks model describes the journey of a user which is conductive and forms a strong habit. It starts with prompting the user just so that the user can take action. It is based on four stages cycle which are trigger, action, variable rewards and investment. Likhachyova et al. (2023), in a paper, examined two different common social mobile applications Uber Taxi and Instagram using the model. Their purpose was to understand the habit of formation and engagement. The researchers found similar hook cycle patterns. while these two applications have different features. Whereas Instagram includes activated calls uber uses fewer triggers

to reassure users. By examining two of the platforms this paper shows the two different types of hook cycle patterns that are internal and external (Falcone and Mazzoni, 2016). Continuous cycles through the hooks let the users start associating with internal triggers that become connected with existing behaviours and structures. According to Ouyang et al. (2022), hookup adhesion is a common spectacle that is divided into two stages. Researchers found many of the beneficial sides of the hook model. It helps to cut through commissions. And hook the attention of the customers which is essential for online customers mostly. To apply the model there are different stages like creating trigger moments that can be either external or internal but not both at the same time. then making it easier for the users to act right and have control over it. Ensuring generosity with rewards including social centricity like tribe, hunt, self etc. It works for developing opportunities to invest (Bright, 2021).

Lean UX is a collective design that approaches agile working which focuses only on resolving issues while with less consuming time. This approach is based on the agile method and makes incremental design processes, and the strategy is fixated on how it can save time and get work done in the shortest possible time. It's about the validation of the statements that cause the build of websites (Hannah, 2023b) there are three fundamental stages of processing or applying the Lean UX model to a website Researchers found that creating low fidelity prototype is the first step here to building or designing the website which outlines the backbone and then it measures elements that is required, but it is based on using the hypothesis to check or to evaluate the assumption that comes from the user end. then lastly the research studies end with a lesson learned through analysing and deconstructing the outcomes taken before implementing any other internal changes. According to the data analysis of the study, there are five core principles of lean UX such as working as a team and always acting updated together, solving the problem only which is necessary and ignoring the rest, making a plan and designing by taking opinion from all the team members, practice flexibility with de-empathise deliverables (Gothelf, 2013).

2.7. Website accessibility – WCAG

According to Spina (2019), WCAG 2.1 accessibility is designated for the use of web applications, which is yet again to be organised by the principles. and there are four principles.

Chee and Weaver (2021) in their research stated that the principles are solely developed by the Web Accessibility Initiative (WAI) of the World Wide Web Consortium (W3C). These

principles aim to enhance the convenience of online visual terms and facts for every other person individually. Here the principles provide the basic concepts of Perceivability, Operability, Comprehensibility, and Sturdiness. It ensures the UI user interfaces and information are sufficient enough with proficiency.to make a smooth understandable and flexible media format it is very necessary. Also to eliminate every possible factor that may cause inconvenient problems or occurrences. According to Abuaddous, Jali and Basir (2016), the challenges come from the websites regarding accessibility which include the lack of knowledge practice work experiences or insufficient knowledge, stability, inpatriate model choosing, wrong applications, choosing the wrong design pattern, colour scales WCAG standards sets the best practice along with critical investigation for website designs. developing the plan and evaluation of the newly implemented website and checking the features thoroughly. Also, it recommends the solutions to any problems related to enforcing web conveniences and its regulations so that the challenges can be solved easily. Khan, Iqbal and Bawany (2014) said in their paper that the evolution of the internet throughout the years with new technologies and interfaces is making every website updated and user-friendly. Every developer is in the race to improve and update their website to be more flexible and user-friendly. in their paper, they investigated the challenges even for the challenged people. with accessibility issues.

2.8. Literature summary

According to the overall analysis from the evolution section here in this chapter, it is clear that many intelligent minds have worked on researching just to find out the areas to improve websites through different approaches. design guidelines are almost the same but the all of the guidelines don’t follow every aspect of design sectors. Like different frameworks, there are different guidelines for UX as well excluding the Nielsen heuristic guideline (Scholtz, 2022), which covers accessibility, usability, systems of navigation process, and information architectures. These guidelines impact the contents of the website and help to establish understandable user-friendly design languages so that it can be less time-consuming, and increase the value of the website along with client engagements. There are seven different factors to influence user experiences. Apart from the methods, designs, and guidelines, users prefer to buy products or take services from a good User Interface UI. Any company, organisational website can be fixed with their UI by following the principles and key findings from the overall research so that they can enhance the satisfaction rate of User Experience UX (Allanwood and Beare, 2019). A bad UX can be a reason behind losing potential

customers if they don’t find the website helpful instead of facing difficulties, they will even lose their interest to buy anything from the site again.

Overall, the chapter gives a transparent idea to improve the user interface and increase the user experience through the perspective of scholars as they depicted their views on different visual appeal approaches.

3. Research Methodology

Research methodology is a way to execute and explain how a researcher wants to carry out the overall process regarding the topic of research (Snyder, 2019). It can be logically made plan that follows the strategies, to resolve a particular issue. The common types of methodologies are quantitative, qualitative, mixed method or experimental, case study. Through discovering, inspecting, processing and generating the research methodologies dynamics are organised.

Intended for this study, the methodology that goes close to the scenario of Allodi Accordions Ltd. is the Qualitative method, to test through secondary research. This company is a kind of E-commerce website where instrumental products are shown, but orders cannot be placed. This research aims to find the visual flaws, through applying this method along with UX principles to fix the user interface so that it can have a better UX.

3.1. Research Philosophy

Research philosophy is a vast area, it is a belief in how data are collected, analysed and formed to use for different statistics. There are four different types of research philosophies such as positivist, interpretivism, pragmatist and realistic research philosophies. This reflects the assumptions and beliefs of the researcher. In retaining any type of research method, some specific steps are crucial for philosophy. It covers the areas of ethics, epistemology, and metaphysics. It deals with the foundations and core development of data.

Allodi Accordions Ltd. is a music instrumental company, which needs a good user interface UI to cope with the bad user experiences. To resolve the issue, the research philosophy which is most relevant to the company situation right now is the positivist philosophy. Apart from all the other philosophies, this is a pattern that counts on dimensions and causes and the knowledge is discovered from unbiased and countable studies. It states if something is not countable, it is certainly unknown (Stephanidis, 2019). It will help to accept the feedback from the company which has already bad experience, and allow to improve the user interfaces.

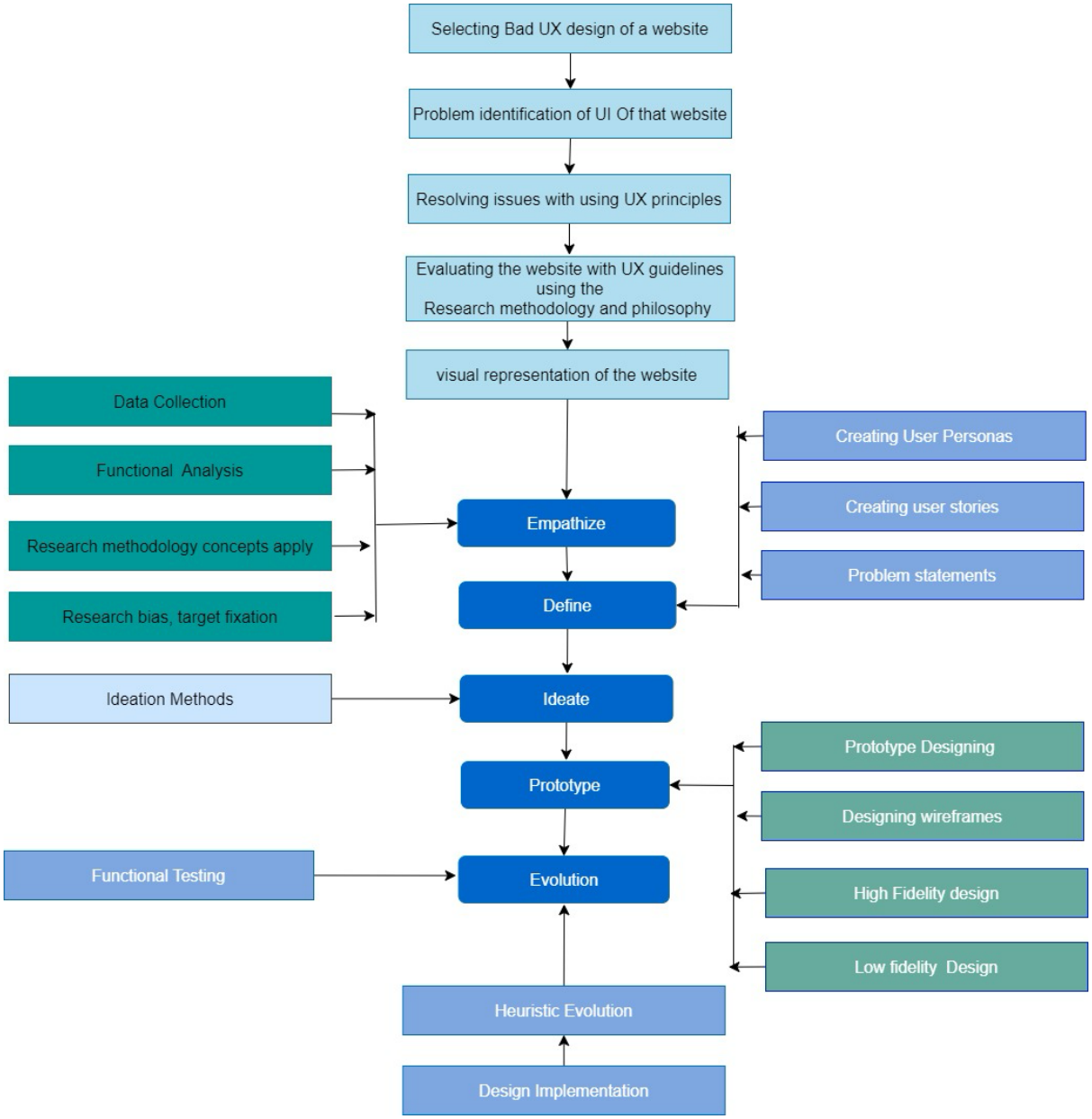
3.2. Research Choice

Among the various types and forms of research philosophies, methods, structures and individuals it refers to the decisions made on the methodologies, frameworks, and techniques. In this research, the researcher focused on the details of the website's usability. It seems to be very challenging to place an order and get to know in detail about the products (Johnson *et*

al., 2013). The website is built on backdated terms. which is very difficult and complex as compared to other currently hyped similar websites. so, to resolve the issues the researcher picked qualitative secondary as the research choice. The reason behind choosing secondary research is that this ensures quality within a short period. Compared to primary research, is very time-consuming and costly whereas secondary research takes less amount of time with lower expenses. On the other hand, the reason behind choosing the qualitative method is, that this is based on theoretical concept and design. Compared to quantitative research which takes numerical data, and is countable or measurable. Whereas, the qualitative process is integration-based and descriptive which aligns with the motives of this study. It is used mostly in organizational bodies, where the source is the Internet. It is a method to be used with the processed data as the data of the website Allodi Accordions Ltd. are already visible, and known.

3.3. Research Strategy

Research strategy is the guideline, a roadmap that guides the research activities to achieve the goals. It helps to clarify the questions regarding the research and recognise the bases while collecting data and resolving the internal issues found in the research. There are a total of five research strategies such as case studies, by taking interviews, or quantitative and action-oriented research (Badke, 2021). In many ways, this research strategies are conducted, which includes the design of the research, its methodology and approach, using longitudinal data, sampling strategies and the techniques of data analysis. It covers three major terms for conducting research which are respectively planning then executing the plan accordingly and monitoring the progress. It mainly prioritizes step-by-step processes at the time of research. But it depends on the researchers, and how they follow the follow rules and strategies to execute (Harris, 2019). It gets easy to execute and track the progress.



Flowcahart1: Step-by-step planning strategy for the research (self-made)

Allodi Accordions Ltd. is one such company with an extremely bad user experience because of its website design. Picking this company and resolving its internal issues using UX guidelines to improve its UI is the research strategy of the researcher of this study. There are no alternatives for a good UI to hook the client's attention but as per this company's website, clients may get confused about buying stuff from them. As it is not at all user-friendly to them, the strategy here is to thoroughly change the features that are becoming an obstacle for the company and the flowchart [1], shows the five stages of guidelines for taking the initiative to implement the website design (Doncaster, 2014).

3.4. Research Bias

Throughout the time of research or this systematic investigation, it has a great chance to influence the other research processes intentionally or unintentionally, it includes the samplings or the processes taken for evolution, and testing that encourage the other existing research results Suter (2012). This unwanted occurrence is known as research bias. and it can occur anytime during the time of doing the research. The researcher maintained the conventional bias during the research and implementation by including misclassified data from other sources and platforms and observer bias, verifying more data sources from competitive platforms. However, managing the research bias underpins the numerous ethics and values to comportment a good paper. This is the most happening reason behind research investigation from its result to alter the findings. But it also results in some poor cogency of research.

3.5. Time Horizon

	W1	W 2	W3	W4	W5	W6	W7	W8	W9	W10
Research & Analysis										
Problem Identifying										
Benchmark Analysis										
Issue Resolving Areas										
Methodology Apply										
User Interface Design										
Low Fidelity										
High Fidelity										
Website UI										
Prototyping										
User Centric Design										
Data Collection										
User Persona Analysis										
Heuristic Evolution										

Design Implementation										

Table 3: Time frame for the research(self-made)

This time frame shows the overall estimated time bound for the conducted research. The steps are sequentially set one by one. the time is set for ten weeks in a row. starting from analysis and research to design implementation after using UX principles and the decision made in this report (Sciore, 2020).

3.6. Selected UX Design

Through UX design businesses can improve loyalty of customers and retention. A product with good UX design creates a strong brand reputation. That is why UX is important to any business. For building a good UX the design requires some guidelines. There are a few famous frameworks for designing good UX.

The UX design that is being selected for this research is User Centric Design (USD).

3.7. UX design framework

The design framework helps designers make logical design decisions and ensures the designers create an enjoyable user experience interface. Following a structured approach while creating user-friendly digital products, websites, or applications is identified as a UX design framework (Cwalina and Abrams, 2018). For this research, the design framework that has been followed by the researcher is User Centric Design (UCD).

The user-centric design framework is created to keep the users at the centre of concern. For adopting a user-centred design framework it is necessary to focus on the user category that'll be going to use the product, defining the tasks that the user will have to complete and addressing the user needs.

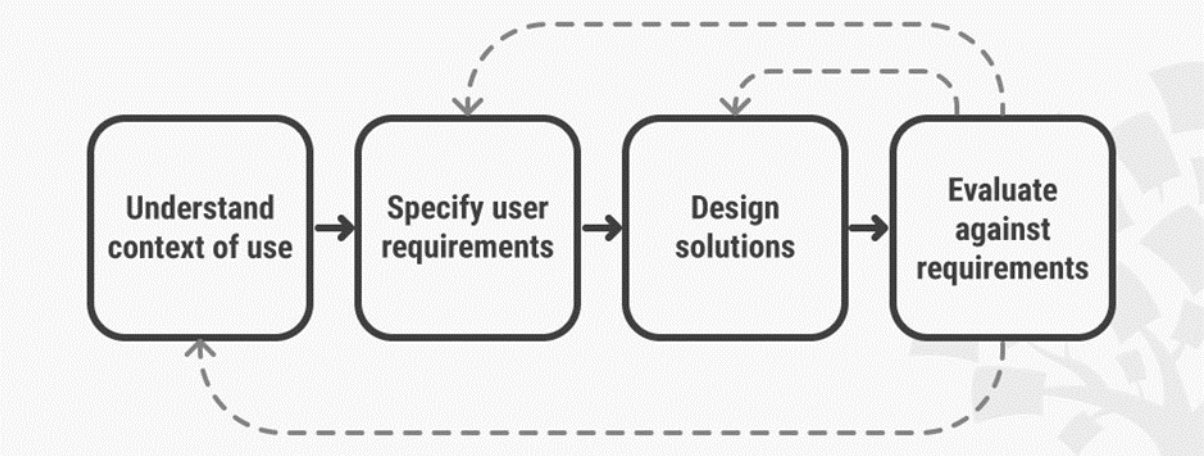


Figure 4: User-centric Desing (Still and Crane, 2017).

The user-cantered framework has four steps:

Understand

In this step designer will have to conduct research and learn about the product that is going to be received by the users. To understand the user needs designer needs to conduct qualitative user interviews, understand user needs, and conduct competitive analysis.

Specify

This step helps the designer to specify the problems that users generally face, working areas to solve the problem, and setting priorities based on resources. Once specifying the problem statement designer can move on to the development process.

Design

On designing process designer needs to concentrate on solutions to the user's problems. In this step, the designer can build user flows, low-fidelity sketches, wireframes, mock-up screens, and prototyping for the betterment of interface design.

Evaluate

In this step designer determines about the fact that which part of the user's problems got solved through the interface design. after the initial usability check the product gets evaluated with real users to gather feedback through usability and interviews.

3.8. Issues

Many types of issues can arise while conducting the research. Mostly the researchers face copyright claims, integrity concerns, and confidential issues. For this research, the researcher has taken every decision carefully. following the copyright laws avoided stealing data from other research results by giving them credit through the citation.to maintain the quality of the research the researcher followed the rules and obligations of research authority. Specifically, these issues are classified as legal issues, social issues and ethical issues.

Legal issues: The researcher followed the standards and rules of legal concerns, considering privacy and security before using any personal or professional data. The researcher was were and aware of all the terms and conditions (Costantini and Forsyth, 2019).

Social issues: These issues include all sorts of social concerns. Diversity in religion, culture, race etc. and the researcher kept in mind while researching so that nobody gets hurt in case of using any personal data.

Ethical Issues: This section works to protect data that can be sensitive. The researcher strictly followed all rules and obligations to prevent the leakage of data information (Pinter and Kuchah, 2021). and used only the relevant information that was needed for the study and made sure no sensitive data was used which can hurt the feelings of someone.

Chapter 4. Analysis and design

4.1. Qualitative Analysis for the target website (http://www.accordions.co.uk)

The qualitative research methodology is used to understand how products, people and things are experienced in the real world. It is a research methodology that is used to Identify and understand people's views, knowledge and actions through gathering non-numerical data and interpreting those data. Qualitative research generates content and provides insight that specifies the research industry (Miles, Huberman and Saldana, 2018). Intended for the Allodi Accordions Ltd UX research determination here the research will be consuming qualitative research methodologies following user interface guidelines for analysing the website's User interface and critiquing its usability experience.

Navigation: After entering the website user is confused about not getting proper information on the home page, this issue creates pain points for the user about their next step.



Figure 5: Inappropriate Navigation menu of Allodi Accordion Ltd's website (Source, Screenshot).

Steadiness: the website didn't follow consistent design elements, it used of same colour and different fonts which created confusion for the users to navigating the website.

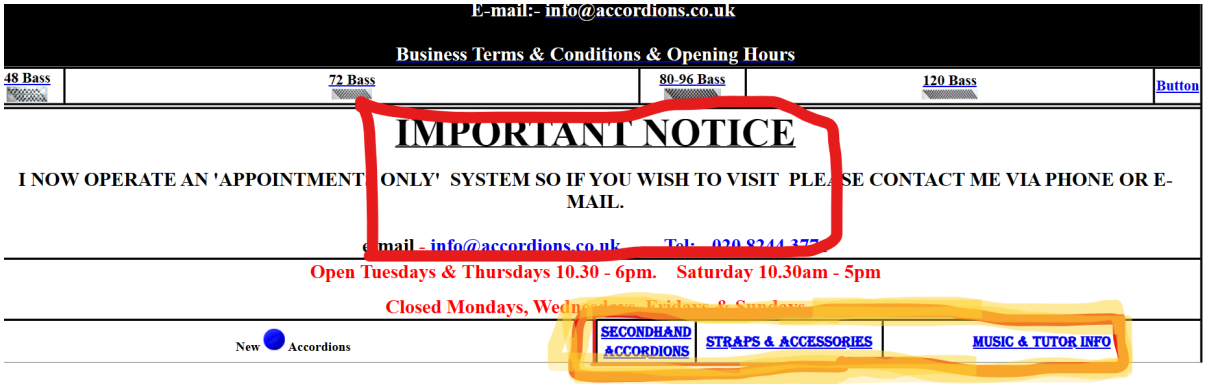
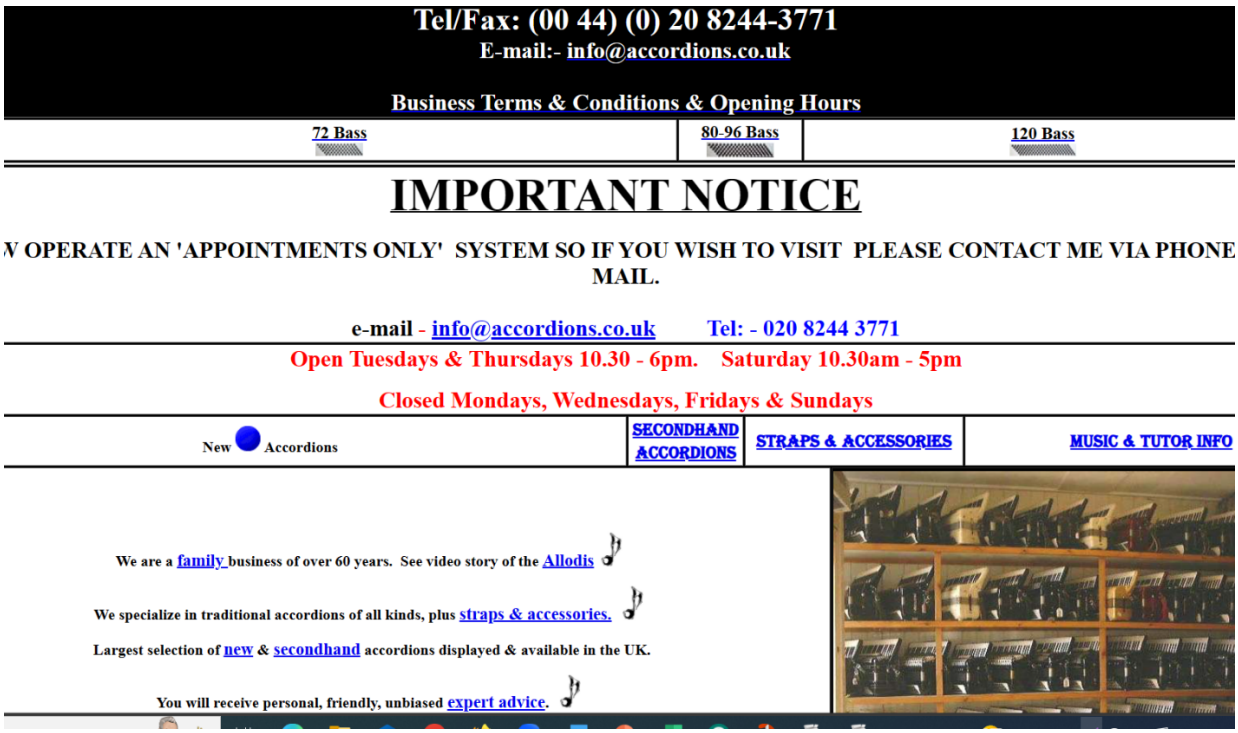
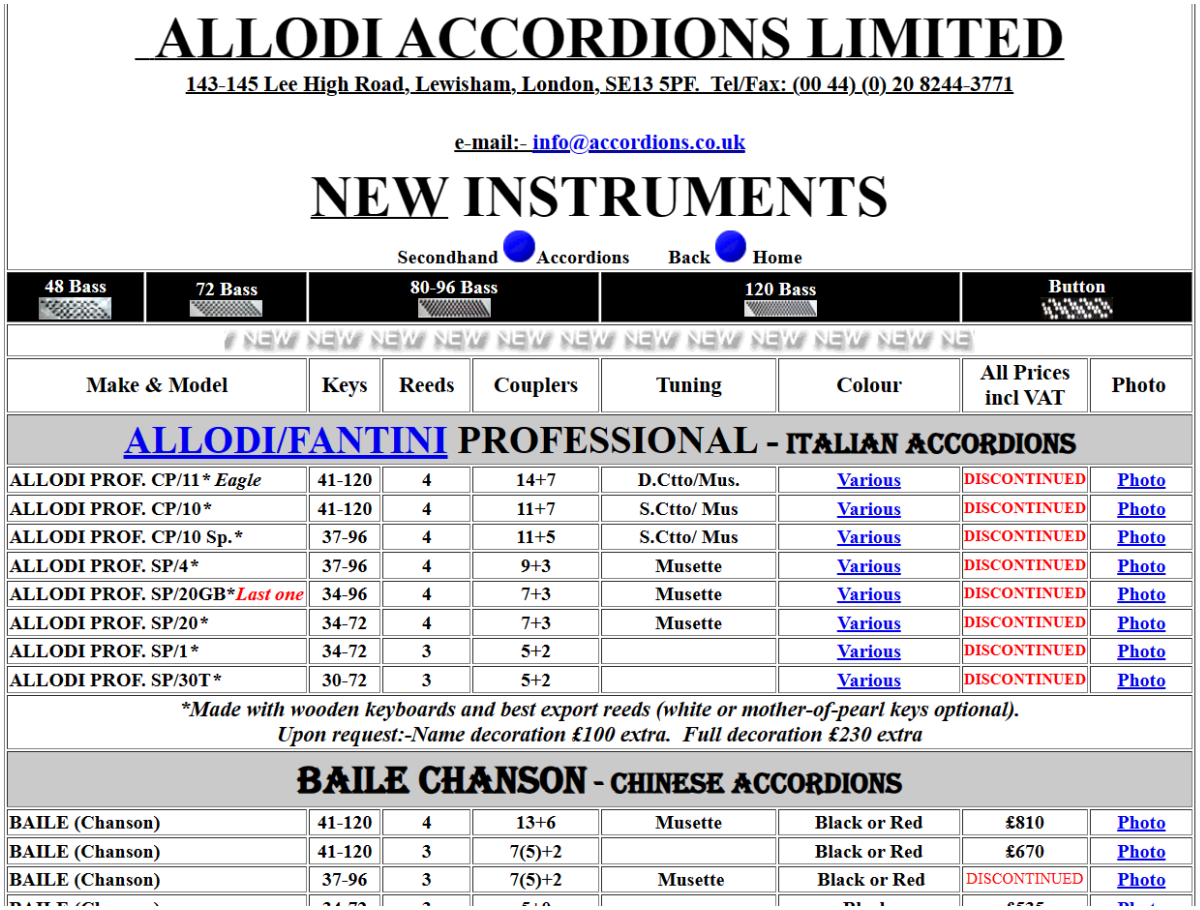


Figure 6: Different types of fonts. (Source, Screenshot)

Colour rules: According to the UIUX guidelines, using too many colours can be the reason for destruction for users. while building any interface it should maintain the rule of using 10% of Primary colour, 30% of Secondary colour and 60% usages of base colour throughout the entire interface. Different colours in the Allodi Accordions Ltd website create a bad user experience for the users. There's no consistency between the colours on the website.





ALLODI ACCORDIONS LIMITED

143-145 Lee High Road, Lewisham, London, SE13 5PF. Tel/Fax: (00 44) (0) 20 8244-3771

e-mail:- info@accordions.co.uk

NEW INSTRUMENTS

Secondhand Accordions Back Home

48 Bass | 72 Bass | 80-96 Bass | 120 Bass | Button

NEW NEW NEW NEW NEW NEW NEW NEW NEW NEW NEW NEW NEW

Make & Model	Keys	Reeds	Couplers	Tuning	Colour	All Prices incl VAT	Photo
ALLODI/FANTINI PROFESSIONAL - ITALIAN ACCORDIONS							
ALLODI PROF. CP/11* *Eagle*	41-120	4	14+7	D.Ctto/Mus.	Various	DISCONTINUED	Photo
ALLODI PROF. CP/10*	41-120	4	11+7	S.Ctto/ Mus	Various	DISCONTINUED	Photo
ALLODI PROF. CP/10 Sp.*	37-96	4	11+5	S.Ctto/ Mus	Various	DISCONTINUED	Photo
ALLODI PROF. SP/4*	37-96	4	9+3	Musette	Various	DISCONTINUED	Photo
ALLODI PROF. SP/20GB**Last one*	34-96	4	7+3	Musette	Various	DISCONTINUED	Photo
ALLODI PROF. SP/20*	34-72	4	7+3	Musette	Various	DISCONTINUED	Photo
ALLODI PROF. SP/1*	34-72	3	5+2		Various	DISCONTINUED	Photo
ALLODI PROF. SP/30T*	30-72	3	5+2		Various	DISCONTINUED	Photo
**Made with wooden keyboards and best export reeds (white or mother-of-pearl keys optional). Upon request:-Name decoration £100 extra. Full decoration £230 extra*							
BAILE CHANSON - CHINESE ACCORDIONS							
BAILE (Chanson)	41-120	4	13+6	Musette	Black or Red	£810	Photo
BAILE (Chanson)	41-120	3	7(5)+2		Black or Red	£670	Photo
BAILE (Chanson)	37-96	3	7(5)+2	Musette	Black or Red	DISCONTINUED	Photo

Figure 7: No colour consistency (Source, Screenshot).

Legibility and Discernability: The website's readability issues were created for not using the proper size fonts and poorly contrasted text. The inconsistency of fonts, size and text style creates readable issues for the users.

Forms and Input Fields: There are no specific input fields for users on the Allodi Accordion Ltd website. By not consuming the input fields it doesn't appraise the clearness of the forms and complicates the users.

Supervision Error: The website doesn't notify or inform about the error with users when the website handles errors. It frustrates the users.

Loading Time: The Allodi Accordions Ltd website doesn't have a long loading time issue. Its processing time is quite smooth.

Visual Hierarchy: The website doesn't rectify its important elements and guidance with proper functional ideas. All the important click points are hardly recognisable on the website's user journey.

User-friendliness: The website's interface is not easily accessible as its design is creating difficulties for users to access the websites.

Call-to-Action: Users face difficulties in finding the Call-to-Action of the websites as users can’t easily identify preferred actions on the website.

4.2. Evaluation of the target website and its competitors using Nielson's Ten Heuristics

Using Jakob Nielsen and Rolf Molich's definition of ten heuristics in interface design is a good practice for gaining usability satisfaction (Hodent, 2017). The generic guidelines in these heuristics have been used for many years for creating websites. Nielsen's ten heuristics will be used here for the Allodi Accordions Ltd website's research purpose.

Visibility of system status

The very first heuristics of Nielsen describes that the visibility of any interface system has to provide information to its users. The website's visibility of Allodi Accordion has not interpreted many error points happening on the screen. Users are getting confused about their next process and that's the reason they have to go through a few extra clicks to get the desired platform.

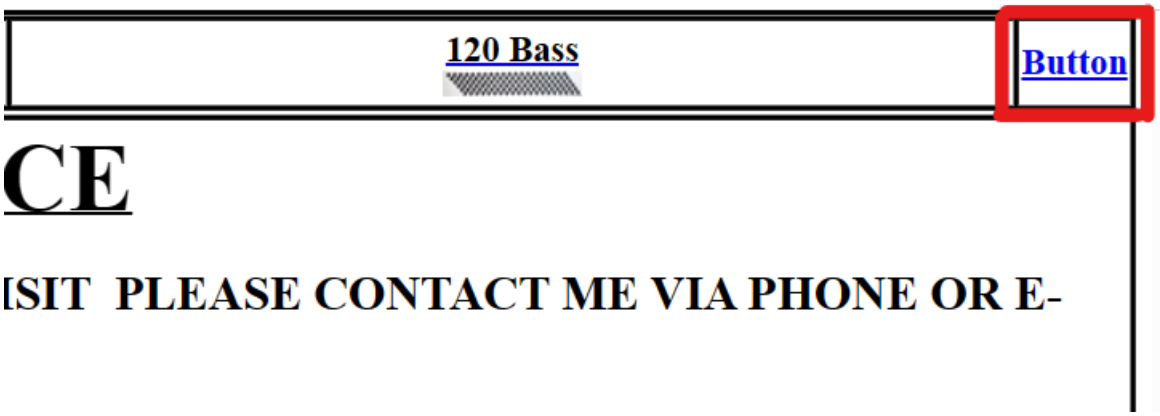


Figure 8: Inappropriate button design (Source, Screenshot)

Identified Issue: while talking about the marked area on the website's landing page, that area is a click point for users to go to the product page. The marked area does not look like a button or a clickable area where the users get confused and it creates a pain point for users.

Match between the system and the real world

The necessity of any website design's language and concept of being familiar to the users and following the conventions of the real world. Neilsen's second heuristics addressed that the information in the interfaces should be presented logically.



Figure 9: Interface issues of Allodi Accordion Ltd. website (Source, Screenshot)

Identified issues: When users enter the Allodi Accordions Ltd website the UI of the website creates confusion for its users. The UI of the website did not provide proper information to its users. The red marked areas in the pictures are the click points of websites which are barely understandable for the users as it has the same colour, sizes, no shadow and background colour. Then again, the yellow marked area in the picture is not a click point but it also holds the same font and colour as the other click areas.

User Control and Freedom

Users have to get clear pathways around the system along with freedom of access and exit from the website and pages in between them. While using the Allodi Accordions Ltd website product information section, after viewing the option there's no 'Exit' to back from that page. Users have to either the search engine's back button to return to the previous page or re-connect with the website. This is a great error in the website's interface which creates a bad user experience for the users.

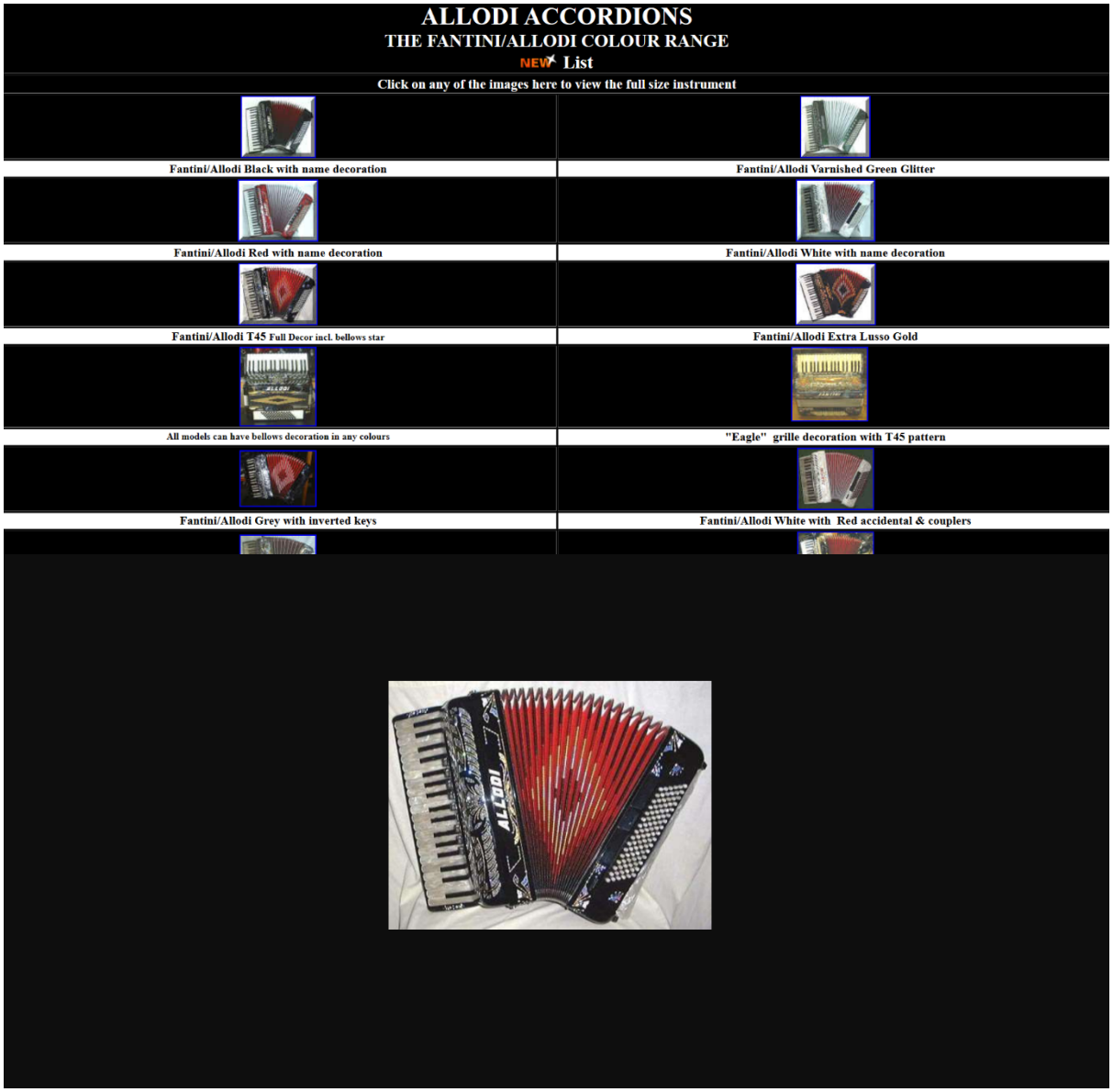


Figure 10: There is no back button option in the picture (Source, Screenshot).

Consistency and Standards

The fourth heuristic, explains that the interface of any design has to maintain a consistent design. Design consistency differs in maintaining a clean and sequent design format so that the whole interface and every page have a similar look and user journey. As one user is using other digital products like **Reverb** that's why user's comfort is to find the same journey that they're used to.

This is the reason users get confused about the journey of the website of Allodi Accordions Ltd as the website UI is almost very different from the other websites that are common to users.

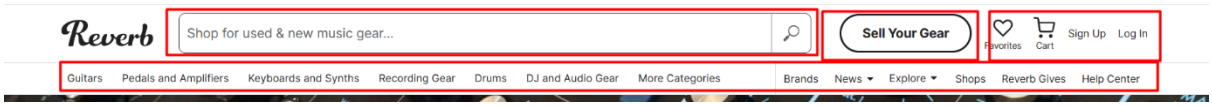


Figure: Heading section of Reverb website (Source, Screenshot).

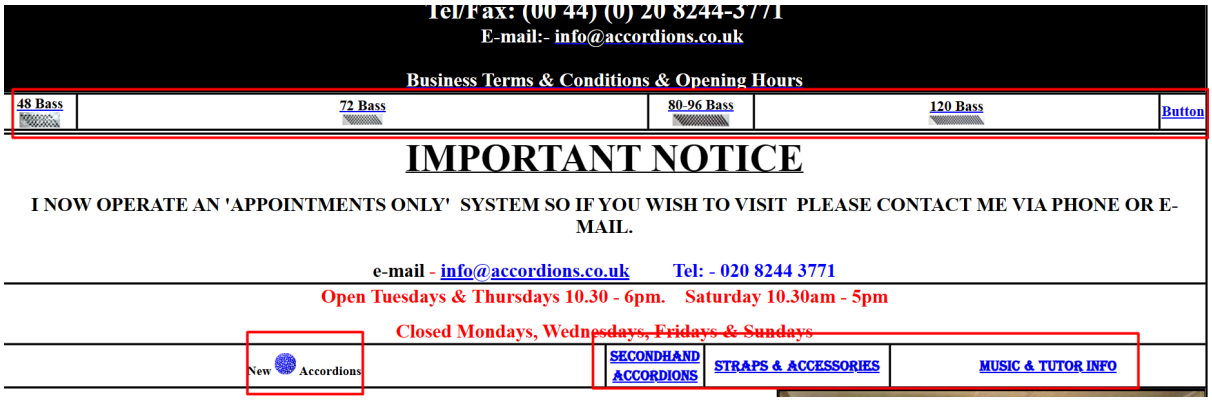


Figure 11: Heading section of Allodi Accordion Ltd website (Source, Screenshot).

Generally, on all the websites there's a dedicated heading section for users to find their necessity from there but there’s no heading section on the Allodi Accordion website.

Error Prevention

The interface should not have many error points in the design. Too many errors make the journey frustrating for users and demotivate them for further use.

One major user requirement on any design is to every page requires a ‘Back’ button to return to the previous page, but there are few pages on the website of Allodi Accordions Ltd. where there are no options to back to the previous page which is a great design mistake.

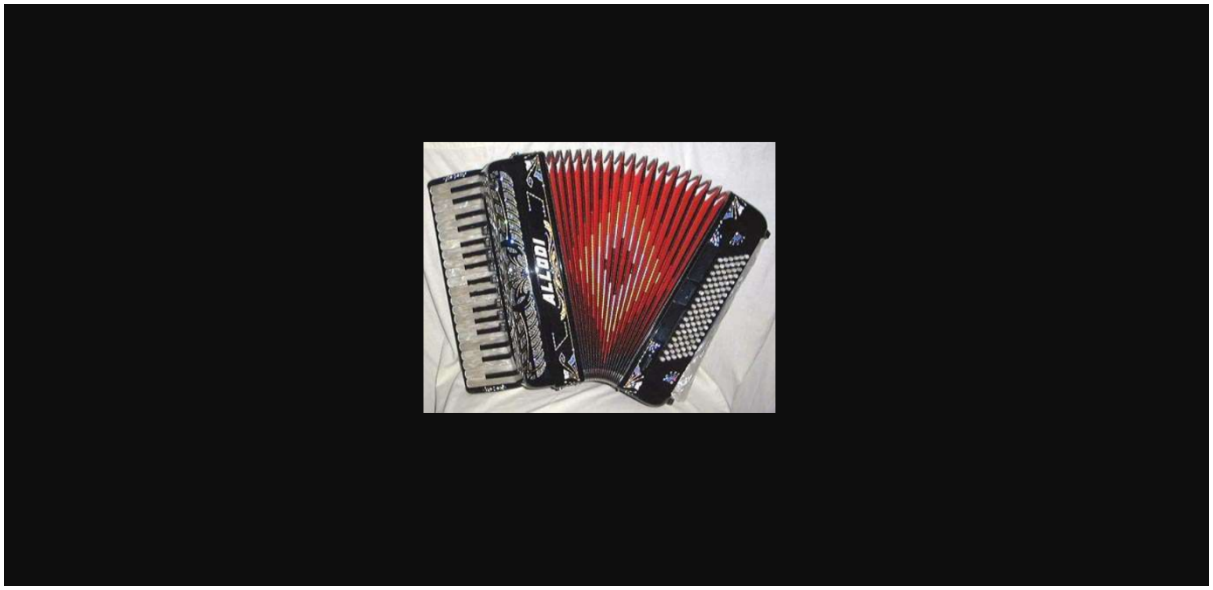


Figure 12: There is no ‘Back’ option on this page of the Allodi Accordion website (Source, Screenshot).



Figure 13: This page of the Reverb website has a ‘Cancel’ option (Source, Screenshot).

Minimize memory Load

It has to be easy to understand for the user for any interface if users get confused and have to analyse every time for the elements of the interface then it will create a bad user experience.

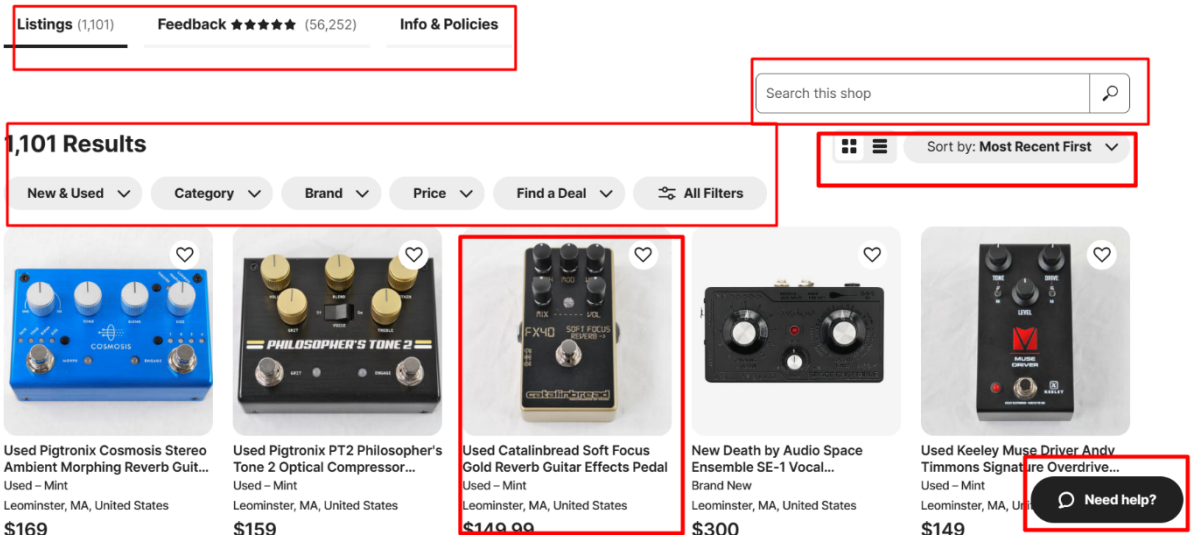


Figure 14: Product page of the Reverb website (Source, Screenshot).

Allodi Accordions Ltd's website doesn't identify proper information with the interface so users have to go through pain points. The product page does not show any pictures of the product on that page, the user has to do some extra click to know the product's details.

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NEW INSTRUMENTS

Secondhand Accordions Back Home

48 Bass | 72 Bass | 80-96 Bass | 120 Bass | Button

NEW NEW NEW NEW NEW NEW NEW NEW NEW NEW NEW NEW NEW

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ALLODI PROF. CP/10*	41-120	4	11+7	S.Ctto/ Mus	Various	DISCONTINUED	Photo
ALLODI PROF. CP/10 Sp.*	37-96	4	11+5	S.Ctto/ Mus	Various	DISCONTINUED	Photo
ALLODI PROF. SP/4*	37-96	4	9+3	Musette	Various	DISCONTINUED	Photo
ALLODI PROF. SP/20GB**Last one*	34-96	4	7+3	Musette	Various	DISCONTINUED	Photo
ALLODI PROF. SP/20*	34-72	4	7+3	Musette	Various	DISCONTINUED	Photo
ALLODI PROF. SP/1*	34-72	3	5+2		Various	DISCONTINUED	Photo
ALLODI PROF. SP/30T*	30-72	3	5+2		Various	DISCONTINUED	Photo
**Made with wooden keyboards and best export reeds (white or mother-of-pearl keys optional). Upon request:-Name decoration £100 extra. Full decoration £230 extra*							
BAILE CHANSON - CHINESE ACCORDIONS							
BAILE (Chanson)	41-120	4	13+6	Musette	Black or Red	£810	Photo
BAILE (Chanson)	41-120	3	7(5)+2		Black or Red	£670	Photo
BAILE (Chanson)	37-96	3	7(5)+2	Musette	Black or Red	DISCONTINUED	Photo

Figure 16: Product page of the Allodi Accordion Ltd's website (Source, Screenshot).

Flexibility and Efficiency of Use

In an e-commerce website users like to get the opportunity to find a specific product in multiple ways. For that reason, user interfaces should be designed with 'Sort' options so that user can set their preferred views. It will give the user a hassle-free experience of online shopping.

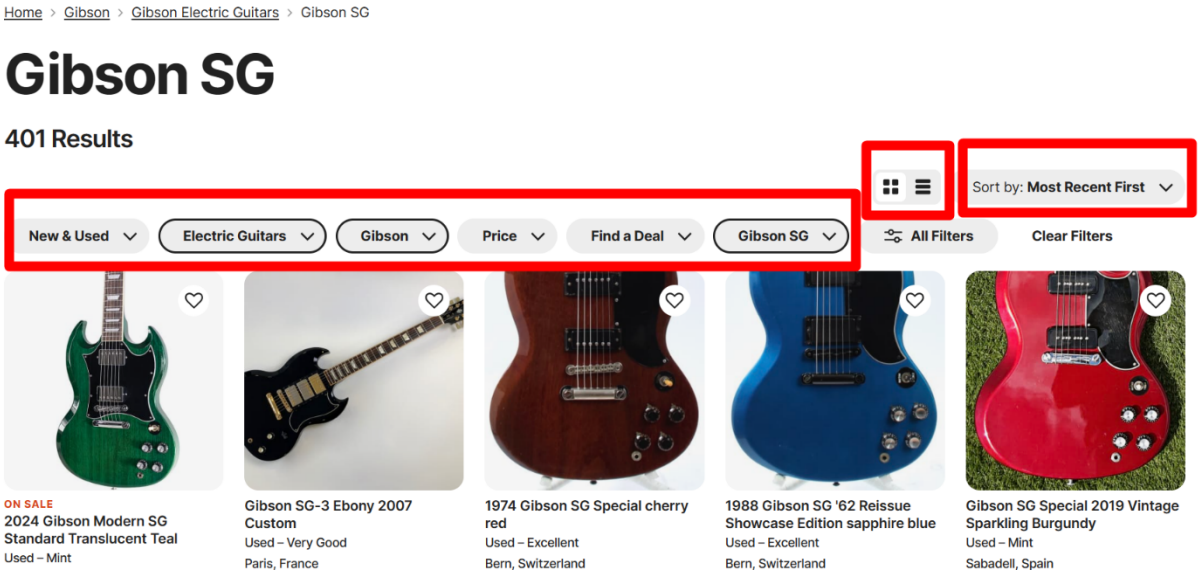
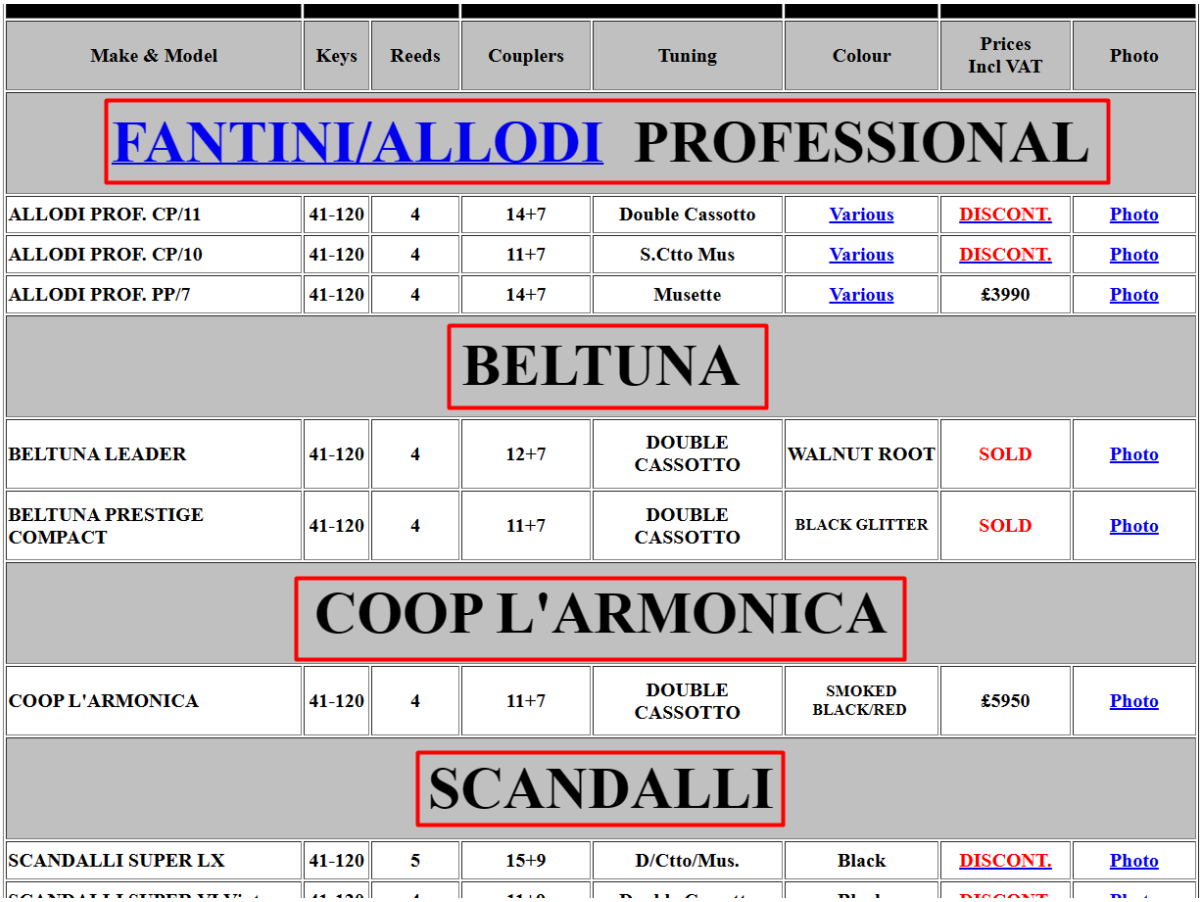


Figure 17: Product 'Sort' option of the Reverb website (Source, Screenshot).

There are no such options on the Allodi Accordion Ltd's website.

Aesthetic and Minimalist Design

The aesthetic design of an interface identifies how the design looks, provides information from a user's viewsight, and a focusable informative design that is easy to understand for its users. Irrelevant information and design should rarely be used on the interface. Showing or Highlighting information that is only necessary for the actions that should be carried out for users.



Make & Model	Keys	Reeds	Couplers	Tuning	Colour	Prices Incl VAT	Photo
FANTINI/ALLODI PROFESSIONAL							
ALLODI PROF. CP/11	41-120	4	14+7	Double Cassotto	Various	DISCONT.	Photo
ALLODI PROF. CP/10	41-120	4	11+7	S.Ctto Mus	Various	DISCONT.	Photo
ALLODI PROF. PP/7	41-120	4	14+7	Musette	Various	£3990	Photo
BELTUNA							
BELTUNA LEADER	41-120	4	12+7	DOUBLE CASSOTTO	WALNUT ROOT	SOLD	Photo
BELTUNA PRESTIGE COMPACT	41-120	4	11+7	DOUBLE CASSOTTO	BLACK GLITTER	SOLD	Photo
COOP L'ARMONICA							
COOP L'ARMONICA	41-120	4	11+7	DOUBLE CASSOTTO	SMOKED BLACK/RED	£5950	Photo
SCANDALLI							
SCANDALLI SUPER LX	41-120	5	15+9	D/Ctto/Mus.	Black	DISCONT.	Photo

Figure 18: unnecessary font size. (Source, Screenshot).

The screenshot addresses the font issue of the Allodi Accordions Ltd's website. When a user lands on this page the first thing that is to take all attention is that marked text which is not necessarily to be in that huge font size.

Help Users Recognize and Recover from Errors

The system needs to guide and inform users about the errors that are happening for the wrong click or input of the wrong information. By informing and guiding system will help the users to solve the problem.

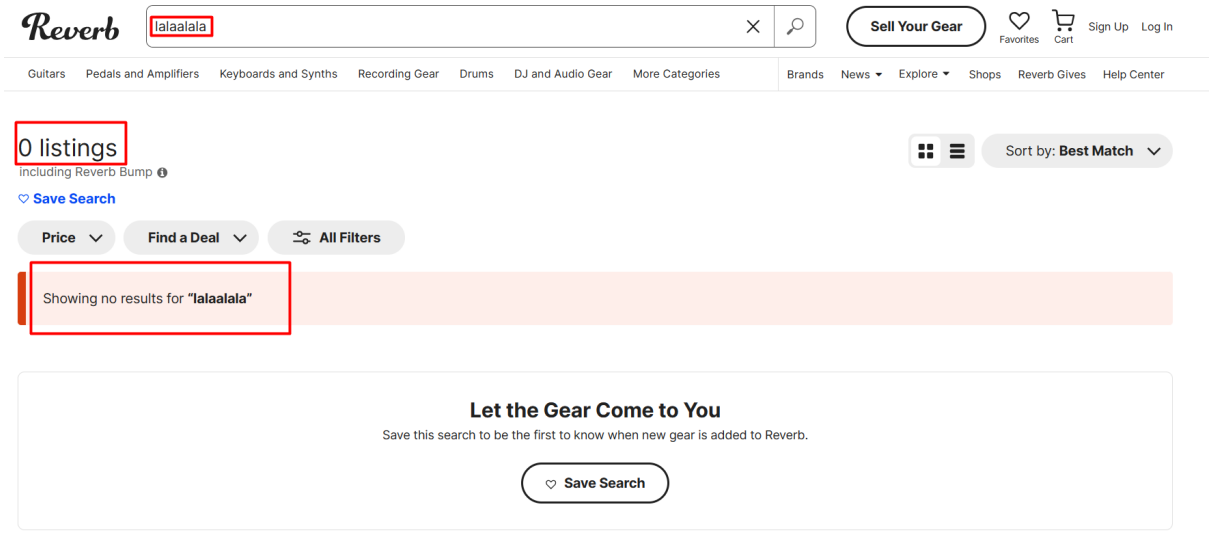


Figure 19: Website notifying error texts for typing wrong text (Source, Screenshot).

The figure shows that if users input the wrong text that is not in the system it shows the error to users acknowledges them and guides them further.

Help and Documentation

Users should be guided with help and documentation so it gets easier for the user to find and focus on the tasks and next steps, it will be time-consuming for the users. This practice builds a good user experience for the website users.

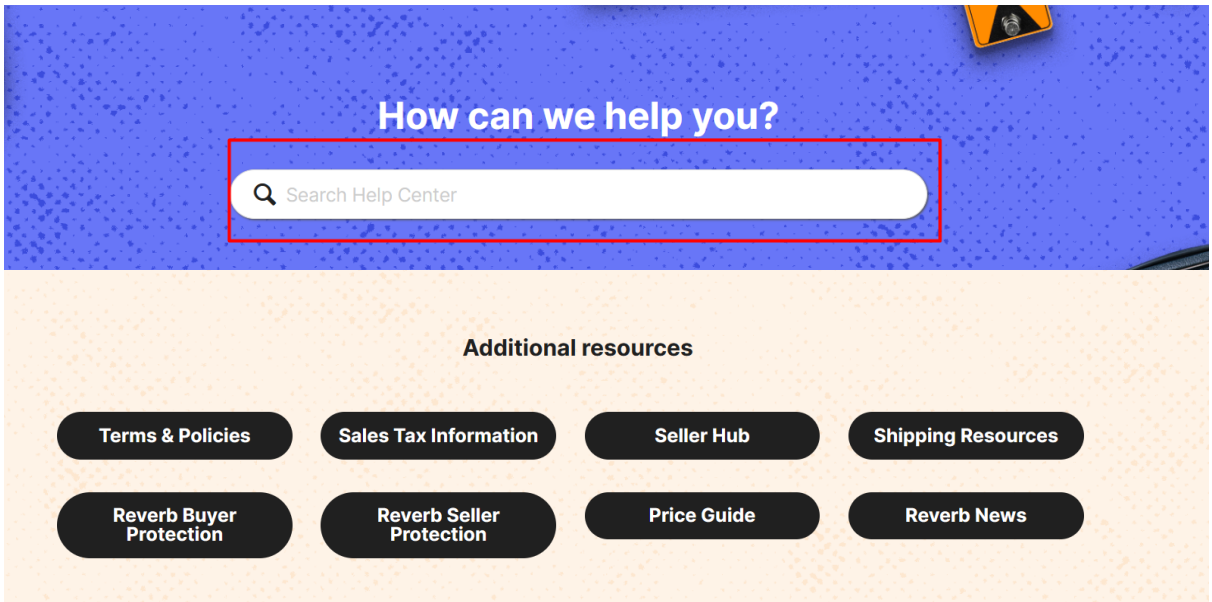


Figure 20: Help and advice option for users on the Reverb website (Source, Screenshot).

In the picture, there's a help-seeking option for users. By using this users will be able to ask for any guidelines and find specific solutions.

4.3. Evaluation of the target website and its competitors using WCAG Analysis

The Web Content Accessibility Guidelines (WCAG) helped developers create accessible websites and web applications. This testing evaluates a website or web application to ensure it meets the WCAG. Here is the UX research journey of Allodi Accordions Ltd. This analysis would be

Using alternate text for images: For making interface content accessible for its users, it is significant to use alternative text or images. Using alternative text on the interface helps the user to get the proper idea even for the content that is on the next page.

Using high contrast text: The website should have sufficient difference to be legible by the visually reduced.

Creating a clear heading hierarchy: The heading hierarchy provides the user clear idea of the content that they are focusing on to meet their needs.

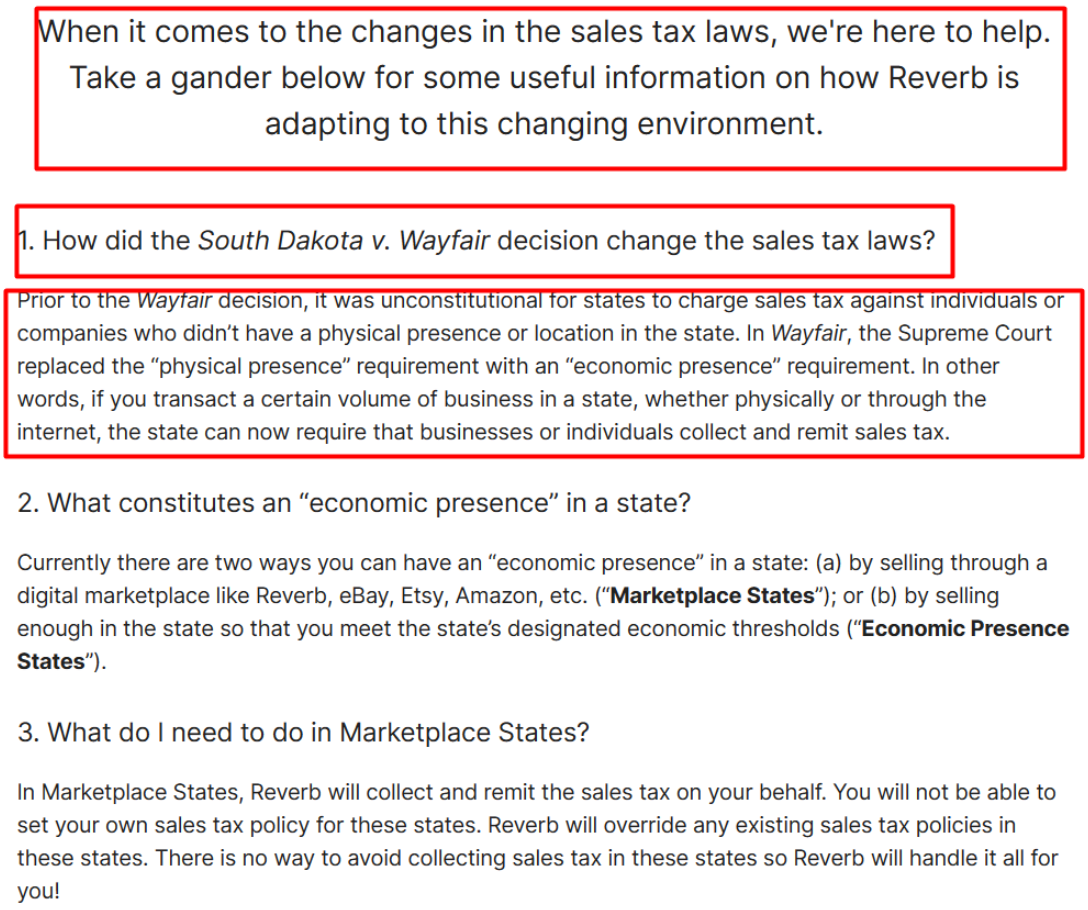


Figure 21: Example of clear heading hierarchy (Source, Screenshot).

The website picture of Reverb shows that maintaining a clear heading hierarchy helps users understand the text properly.

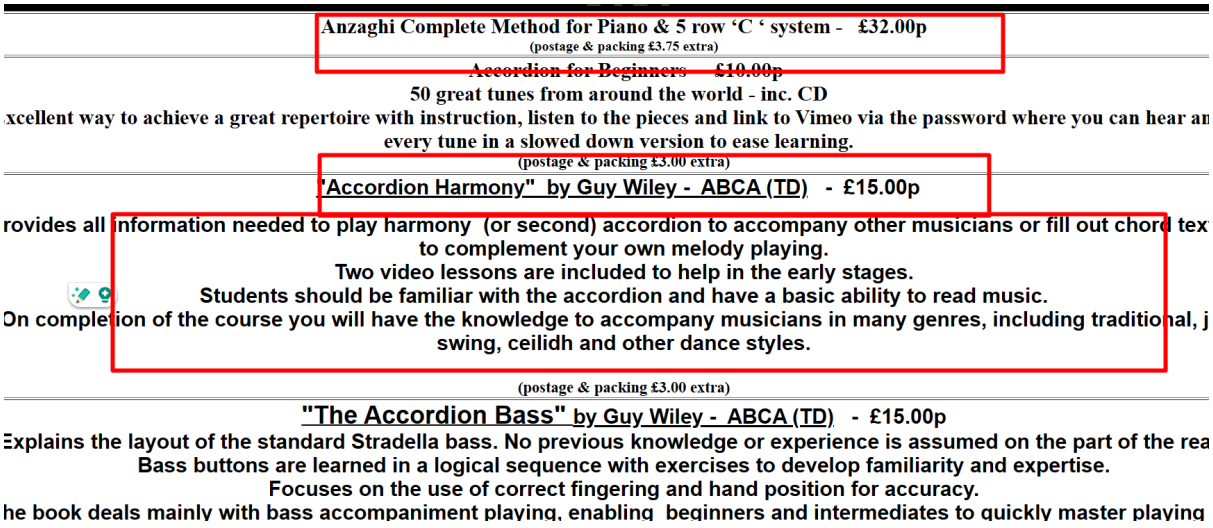


Figure 22: Example picture of not clear heading hierarchy (Source, Screenshot)

The screenshot of Allodi Accordion Ltd's website shows that it didn't maintain any clear heading hierarchy that's why it is quite difficult to understand the key writing of that segment.

A limited number of navigation links: On the navigation menu the preferable links are four to five. The navigation menu is very important for any system so a maximum of seven primary links should conducted on the navigation menu.

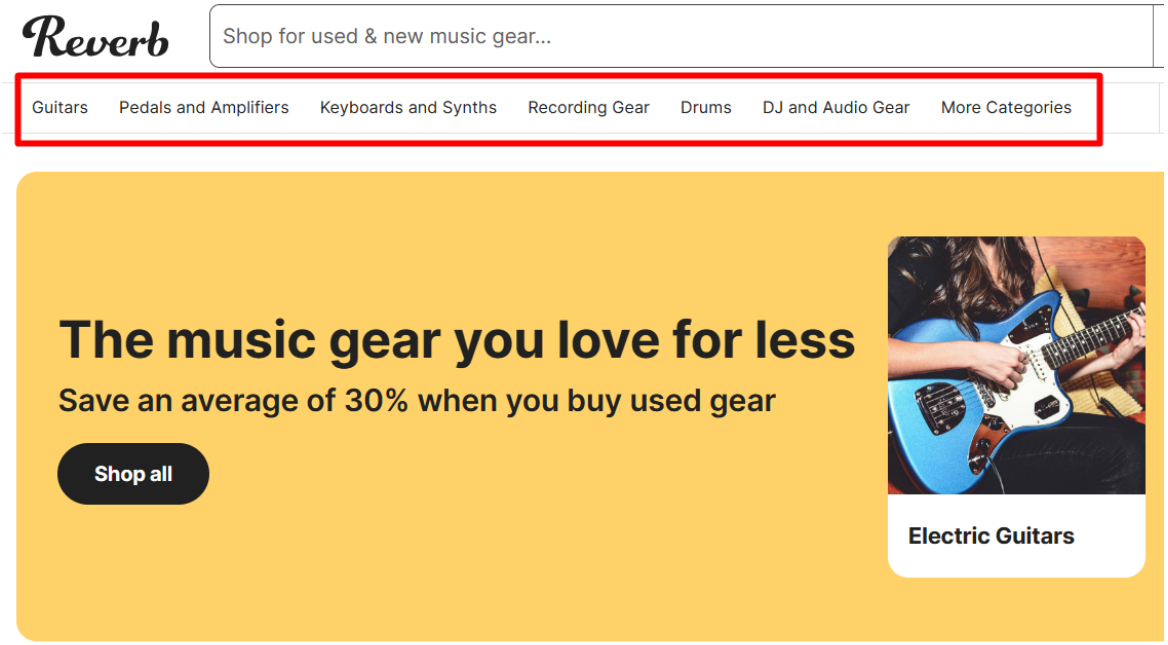


Figure 23: Example of Ideal navigation bar (Source, screenshot).



Figure 24: Navigation area of Allodi Accordions Ltd's website (Source, Screenshot).

The picture of the navigation area of the Allodi website, it is showing that there is no such ideal navigation area in the website. That is why the users of this website are having difficulties finding accurate information.

Creating clear and concise labels: Labels of forms are very important for any web system's wireframe involving user information input like signup, registration and contact details or feedback forms. This helps better understand users and their needs.

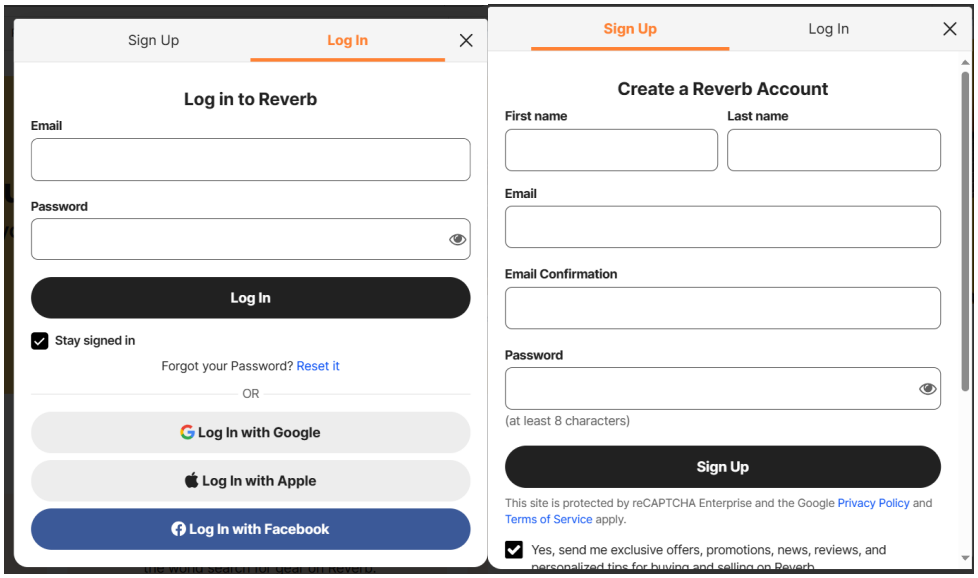


Figure 25: Information forms (Source, Screenshot).

As an example of clear and concise labels here are using the information forms of the website Reverb.

But there's no such thing as an information form on the website Allodi Accordion Ltd.

Chapter 5. Implementation and testing

5.1. Understanding the user and context of use

User experience design describes the whole experience regarding a service or product, an enchanting account of the user's expressive state, the situation of use or atmosphere, and the designed system or artefact. Using of the context describes the conditions and environment under which a product or design will be used.

5.1.1. User Research

For re-designing the Allodi Accordions Ltd. Website user research is important to understand the problem of the users. User research will identify the exact user problem that needs to be

solved through redesign. It is important for the researcher not to come to any conclusion from self-realization the researcher has to identify the user's problems, pain points, needs, desires, feelings and behaviours.

User Research Objectives

- **Purpose-** The purpose of re-designing Allodi Accordions Ltd.'s Website is to provide a better user experience by solving the current user issues of the website.
- **Knowledge gaps-** The knowledge gap of the current website is that it didn't follow many user guidelines such as **visibility of system status**, **match between the system with the real world**, **user control and freedom**, **consistency and standards**, **error prevention**, and **efficiency of use** while building the website.
- **Functionality-** Though the accessible time of the website is quite fast it is not satisfying the users with the interface experience.
- **Problem fixing area-** There are quite a few problems with websites such as the website didn't follow any method during its building process this is the reason the website is not captivating with proper information for its users, the colour usage of the website wasn't followed the colour guidelines, Users has very less freedom and decision making opportunities on the system, user getting confused while accessing the pages of the website because of there's no proper usage of heading and specific navigation link for users. For all of these reasons, users get confused while navigating through the websites.
- **Proposed area-** This research will bring a clear identification of the website user's pain points that are created because of a bad user experience with the website.

Research Identification

- **Website type:** E-commerce
- **Products and services: both** New and second-hand Accordions. Piano Accordions. Accordion repairing service provider.
- **Visitors' area of interest-** The visitors to the website are mostly people who have a passion for musical accordions, Tropipop Musical Artists, and rock musicians.
- **Age-** The age of the visitors is mostly from teens to late seventies.
- **Target audience:** As an Accordion is an old musical instrument most of its users are adults who have an interest in playing Accordions, the visitors to the website are

mostly adults, and some of them might have issues with colour blindness, low eye sights, observing issues etc.

5.1.2. Persona Development

A user persona is important for understanding users' motivations, needs, and pain points. User persona identifies the target user of a product, service, or website. User persona is being created based on research on user data. To understand the business audience and design needs, a user persona is being created. For identifying target users and their needs from the Allodi Accordion website the research followed user persona creation. The target users' personals are-

- Young Adults, people over 25 years
- Emotionally dependent on music
- Passionately follow the musical instrument industry

User Persona 1: Young Adults, people over 25 years



User Persona 2: People's passionately follow the musical instrument industry



James Thompson
Software Engineer

A 50-year-old man who enjoys playing piano

james.thompson@gmail.com

About

50 Years Old

Man

London, England

Goals

- Plans to enroll in piano masterclass led by a renowned pianist.
- Establishing a free weekly piano workshop for underprivileged children in his neighborhood.

Frustration

- Limited time for piano due to demanding software engineering career

Biography

a 50-year-old software engineer from London, finds solace and passion in the piano. Despite a demanding career, he dedicates time to composing and attending concerts. Music is his lifelong companion, guiding his emotions and experiences. James shares his love by teaching piano to eager students, believing in its power to inspire and connect people.

Personality

Communication 74%

Analytical 73%

Problem Solving 76%

User Persona 3: People emotionally dependent on music



5.1.3 Empathy Building/ User Journey Maps

Stage	Awareness	Consideration	Decision		Delivery and use			Loyalty and Advocacy	
User Activities	Hear from friends, see offline or online	Compare and evaluate alternatives	Add to shopping cart	Make an order	Receive or pick up an order	Contact with the shopkeeper	Enjoy Product	Order again/ order more	Share experience
User Goal	No goals at this point	Find the best solution to buy Accordions	Find and select products easily, get inspired	Order effortlessly	Receive or pick up an order effortlessly and when needed	Get help if problems appear	Have the right and good quality	Repeat good customer experience	Share feelings, give feedback
Touchpoints	Musical interest, Instrument Odessite	Word of the mount, physical store, social media	Media Branding	Website app order confirmation email	Phone Email	Packages			
Business	Increase sales, and	Increase the number	Increase shop	Deliver on time and	Increase custo	Match expect	Increase orde	Turn negati	Turn negative

Goal	custom er visit	of website visitors	ping cart value	maintai n the deliver y window	mer service satisfa ction	ation	r valu e	ve	experie nces into positiv e

5.2. Specifying user and business requirements

5.2.1. System requirements

Allodi Accordion Ltd is a family business selling different types of Accordions. To reach out to many customers and for the business's online identification, it built an e-commerce website to grab versatile clients. The system has to be built in a way that users get a good user experience while visiting the website.

5.2.2. User requirements

User requirements for the Allodi Accordion Ltd's website:

User Registration and Verification:

a) Users must be access to register with a unique username, password, and email address.

b) User's accessibility to log in using their registered identifications.

c) Users should be able to recover or reset their password if forgotten.

User Profile Management:

a) Users should be able to view and edit their personal information, such as name, email, phone number, and shipping address.

b) Users should be able to manage their payment methods, including adding, editing, and removing credit/debit cards or other payment options.

Browsing and Searching:

a) Users must be able to browse the available Accordions by category, new product, or second-hand product.

Product Details:

a) Users must be able to assess complete information about an Accordion, with model, size, price, brand, and a brief description.

b) Users should be able to read reviews and ratings from other users for each Accordion.

Shopping Cart and Wishlist:

a) Users must be able to add Accordions to their shopping cart for purchase.

b) Users should be able to save products to a Wishlist for future reference or purchase.

c) Users must be able to view, edit, and remove items from their shopping cart and Wishlist.

Checkout and Payment Option:

a) Users must be able to smoothly complete the checkout process, with entering shipping and payment information.

b) Users should be able to apply discounts or promotional codes if applicable.

c) Users should receive an order confirmation with a description of their purchase, including an order number and expected delivery date.

Order Supervision and Tracking:

a) Users should get access to view their order history, with the status of each order (e.g., processing, shipped, delivered).

Customer Care:

a) Users should have access to customer support through multiple channels, such as email, phone, or live chat.

b) Users should be able to easily find answers to frequently asked questions or access help documentation related to the online bookstore application.

Discounts and Promotions:

a) Users should be able to view and take benefit of any ongoing discounts, promotions, or special offers.

5.3. Create design solutions

5.3.1 User flows

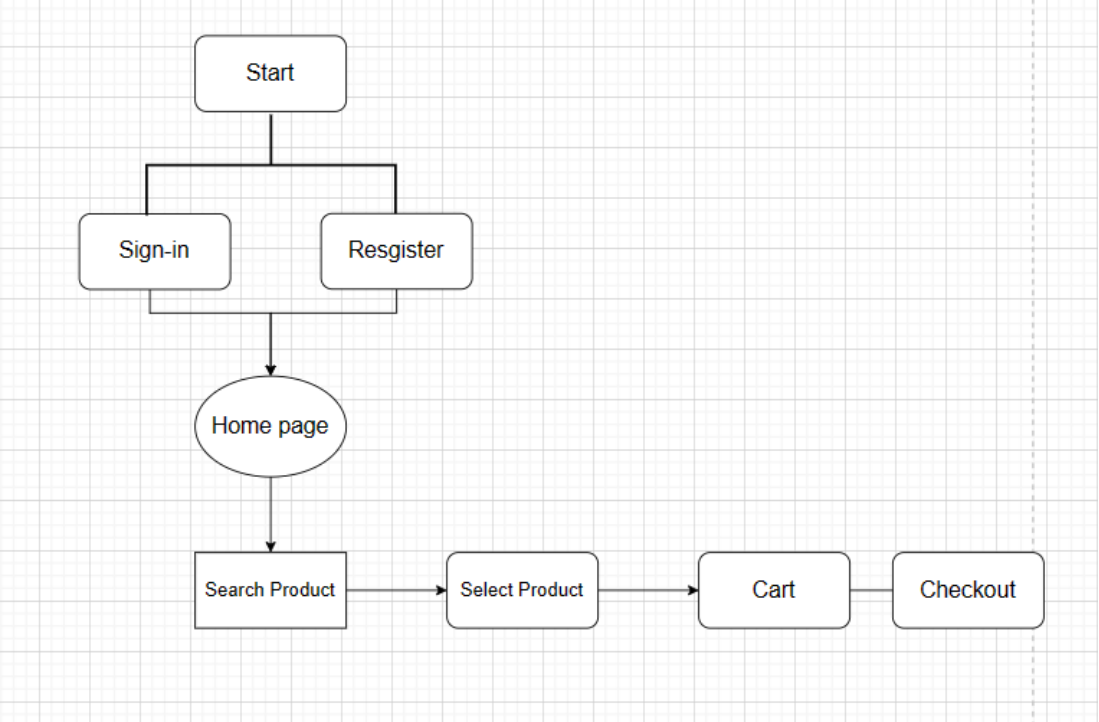


Figure 26: User flow diagram of Allodi Accordion Ltd. Website (Source, self-made).

5.3.2 Colours, icons, images, language, and tone

The advantage of using the right colour combinations in website design it helps users understand and navigate the site. The usage of colours enables users to find information that is needed to respond to a 'Call to Action (CTA)' (Ridgway, 2017). Three colours that are being chosen for the Allodi Accordion Ltd's website re-designing are

Primary Colour: Rosewood is chosen as a primary colour for the website. As per the colour guideline, the primary colour is being used for 10% of the entire website. The primary colour is used on the buttons and on the important 'Call to Action' areas of the website.

Secondary Colour: The secondary colour of the website is 'Almond' which is the lighter shade of the website's primary colour. The secondary colour is being used 30% of the entire website.

Accent Colour: The accent colour is used in 60% of places on the website. The accent colour is used on the maximum of the font's colour.

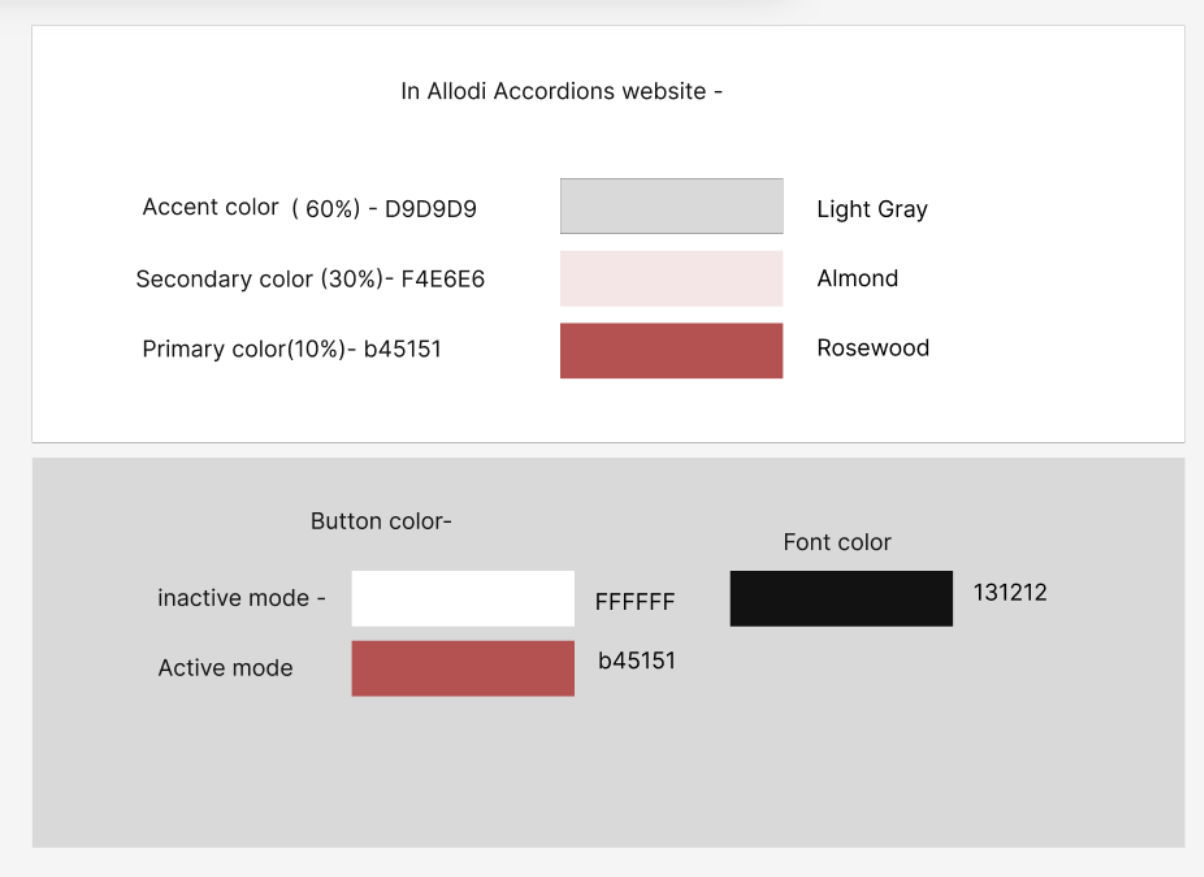


Figure 27: Colour theme of the website's re-design (Source, Self-made).

5.3.3 Wireframing and Prototyping

The process of turning an idea into something perceptible is known as prototyping. The process of prototyping can be established and upgraded (Uklejewski, Rogala and Winiecki, 2023). All the pages of the website re-design have connected with the prototyping mode.

Sign-in page

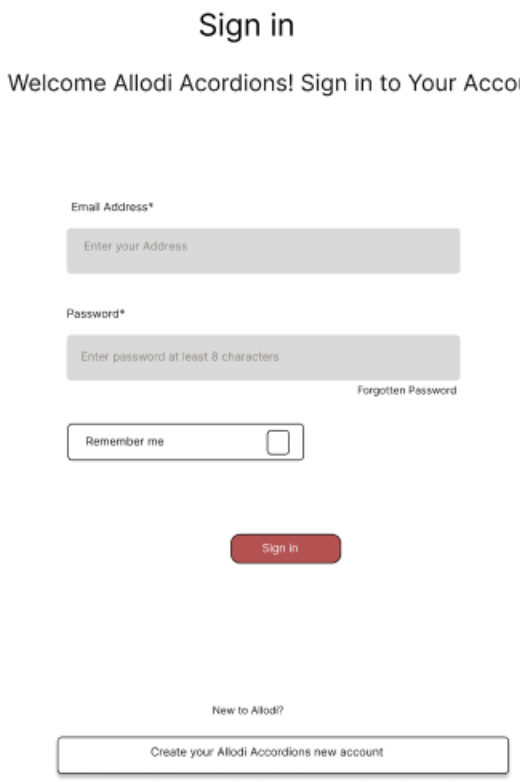


Figure 28: Re-design Allodi Accordion website's sign-in page (Source, self-made).

On this page, the user will be able to complete their sign-in process. To complete the sign-in process user needs to fill in the information box with email and password. After providing the information users are required to click on the sign-in button to complete the sign-in process. This page has the link to registration for the user to go through the registration process.

Navigation-Section

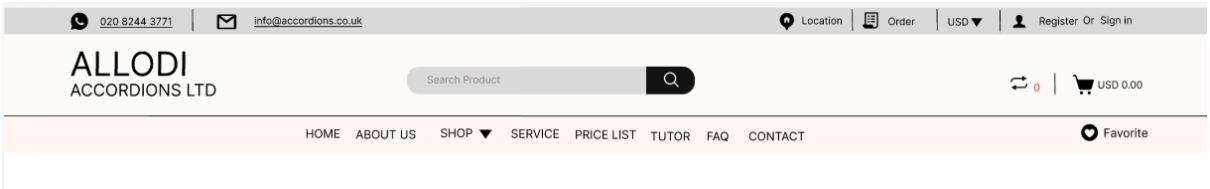


Figure 29: Re-design Allodi Accordion Ltd. Website's Navigation menu (Source, self-made).

The navigation section has links to all core accessible features of the website.

Clicking on any feature will take the user to that specific web page. Also, there is a search engine linked for users to meet their specific requirements.

Product Page

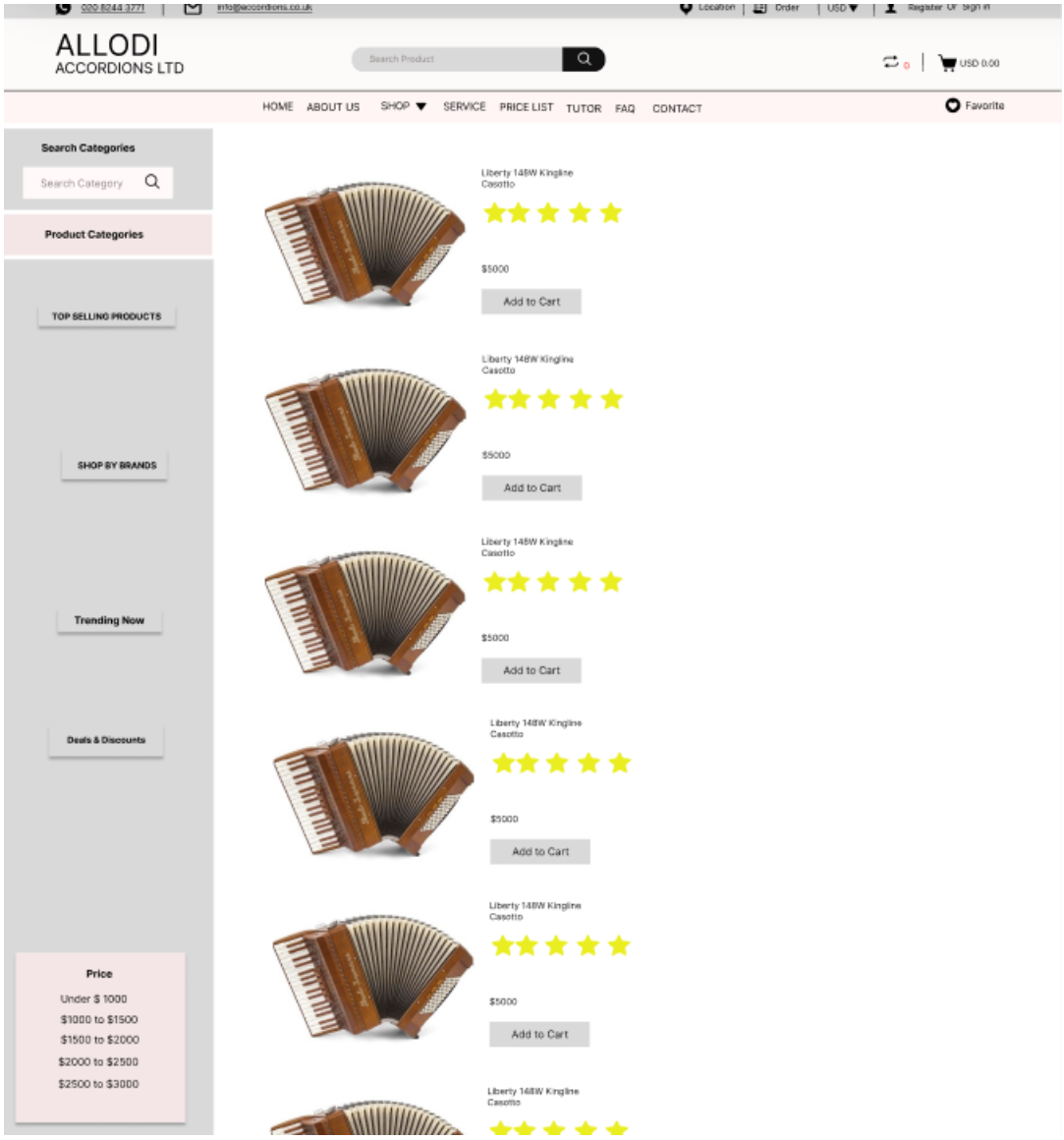


Figure 30: Re-design Allodi Accordion Ltd. Website's product page (Source, self-made).

On the product page user can see the list of products and product details. On this user can add the product to their cart by clicking on the 'Add to Cart' button.

Shopping Cart

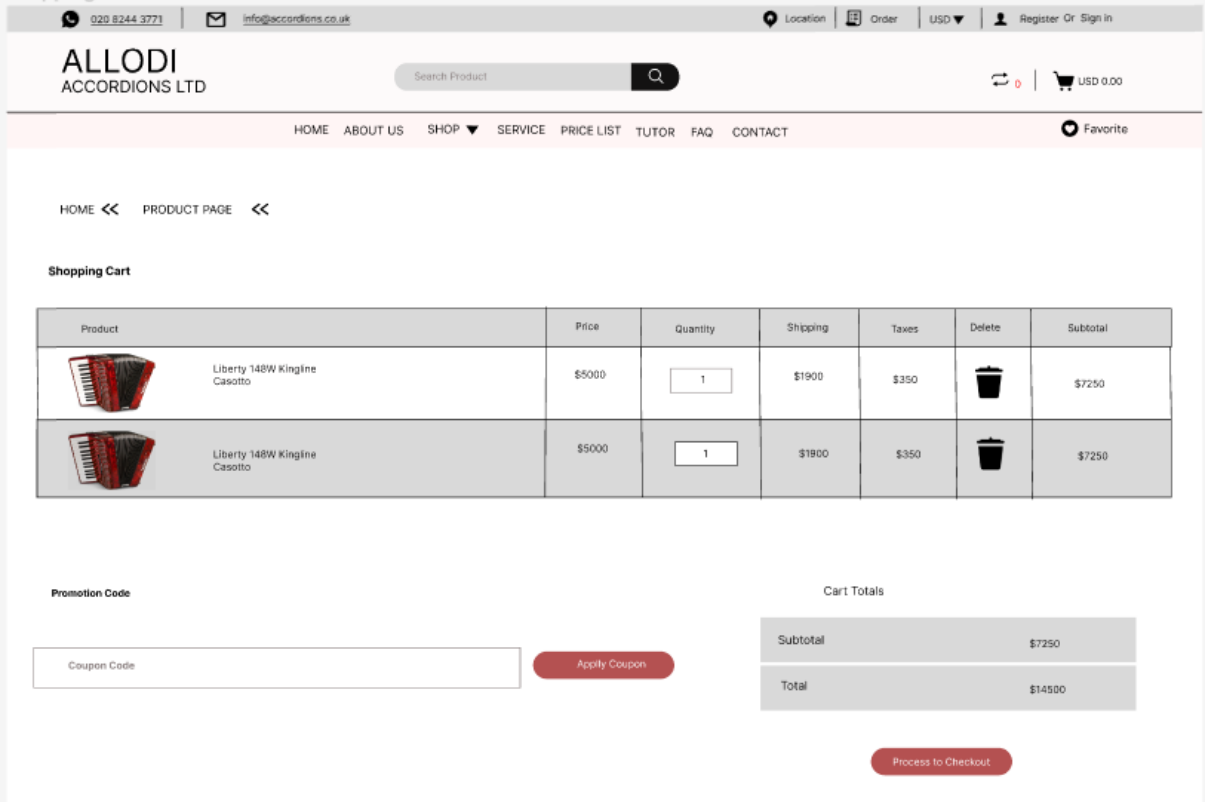


Figure 31: Re-design Allodi Accordion Ltd. Website's Shopping Cart page (Source, self-made).

On this page, users will be able to see their shopping status, details of their chosen product, and the total amount of payment. Users will be able to delete any of the chosen products on this page if they feel the necessity.

Checkout

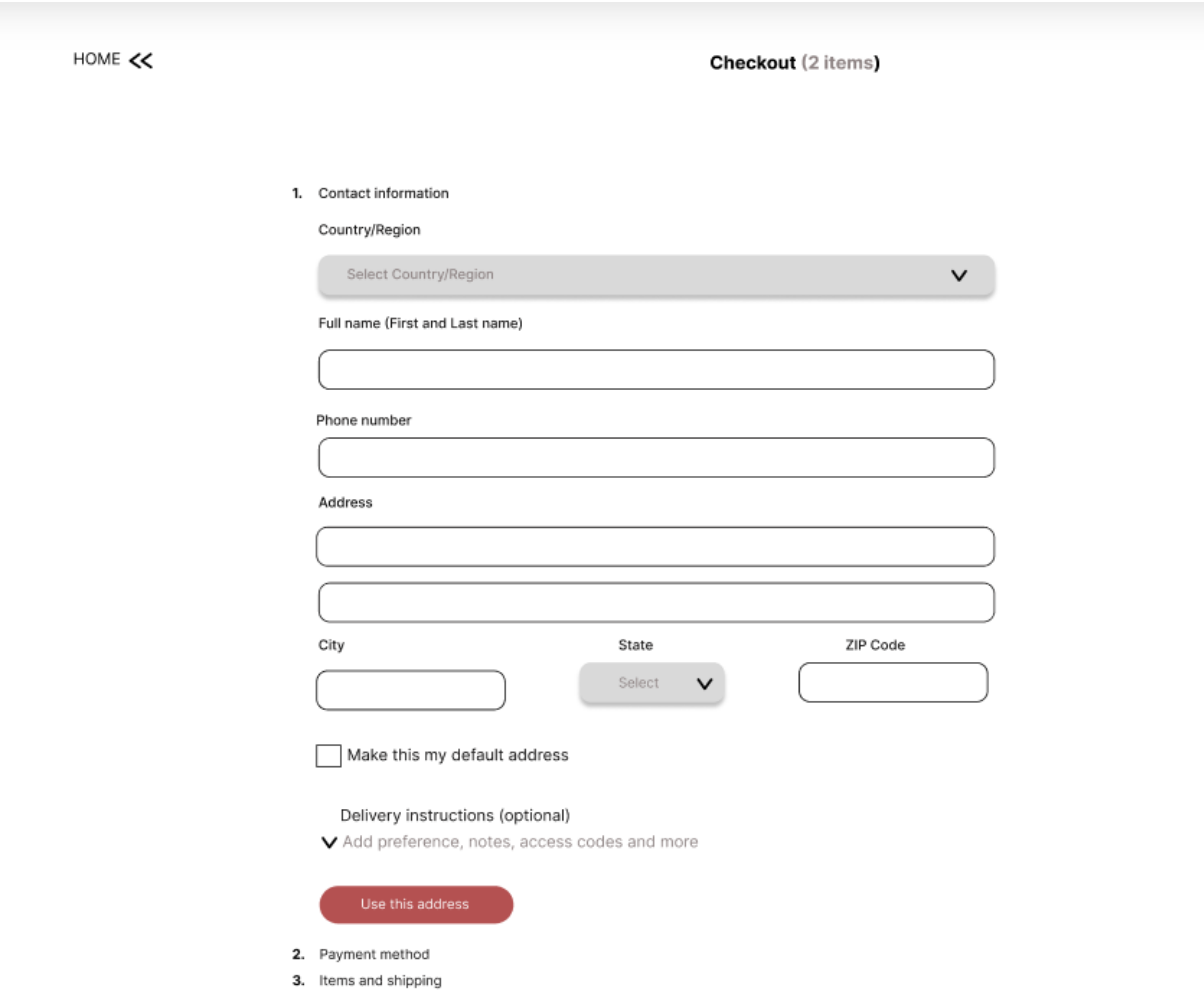
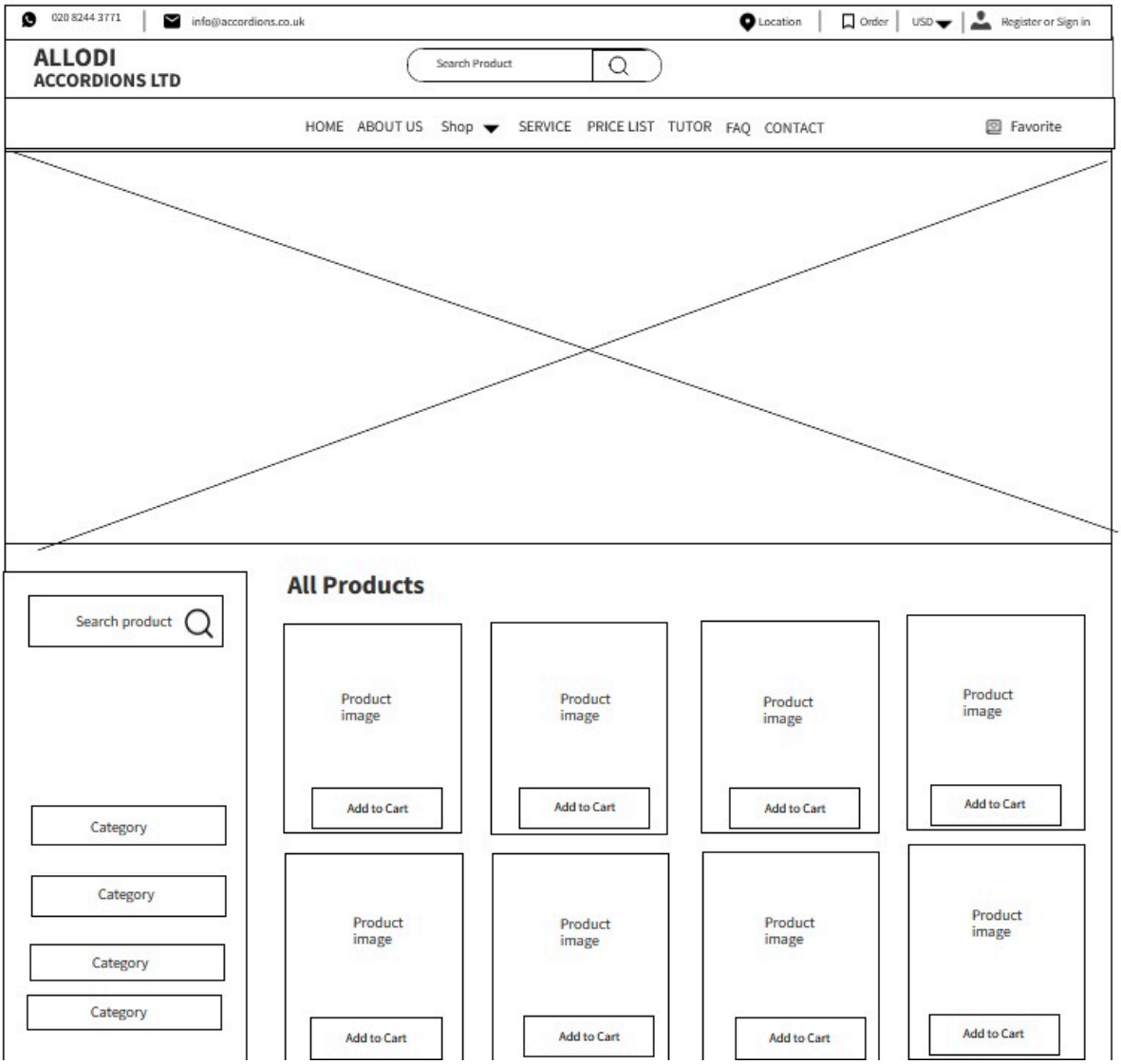


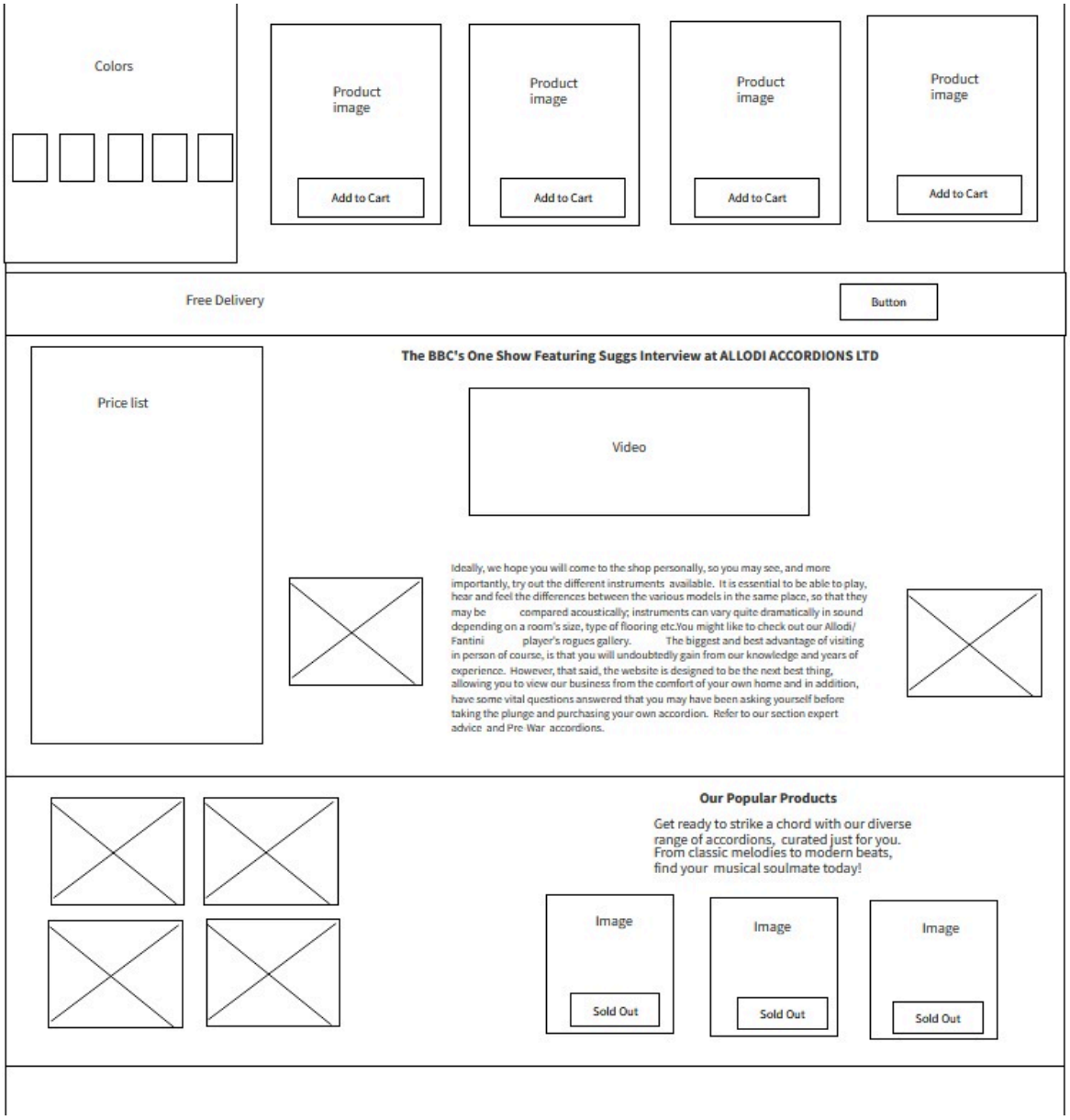
Figure 32: Re-design Allodi Accordion Ltd. Website's Checkout page (Source, self-made).

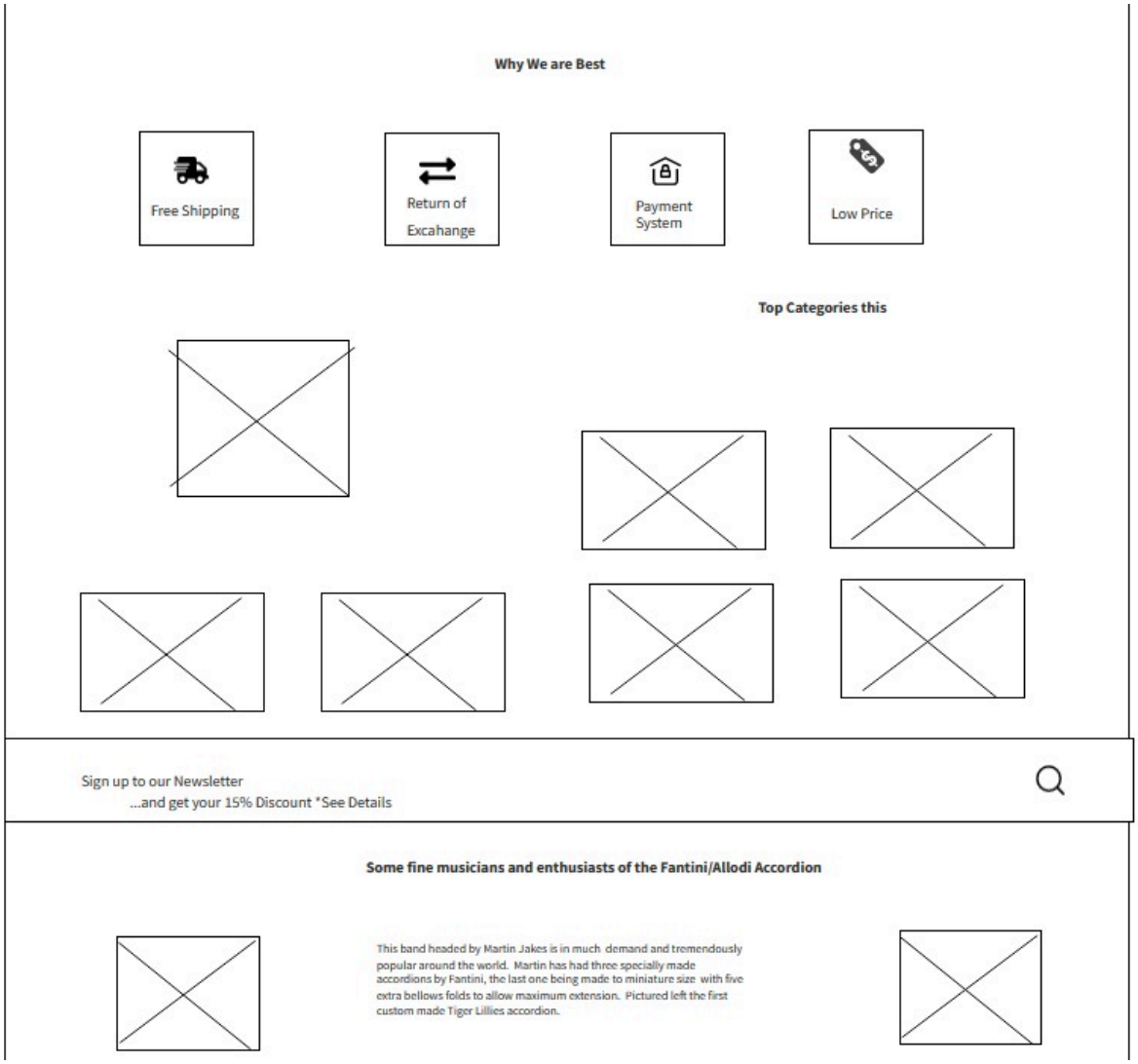
After selecting the product for purchase, on this page, the user has to fill all the information fields with the necessary information to complete the purchase process. The prototype mode was added on the field of 'Country' and 'State' sections, clicking on these sections users will be able to select from the options. After filling up all the fields with appropriate information user has to select their payment method. The 'payment method' page link will be connected to this page.

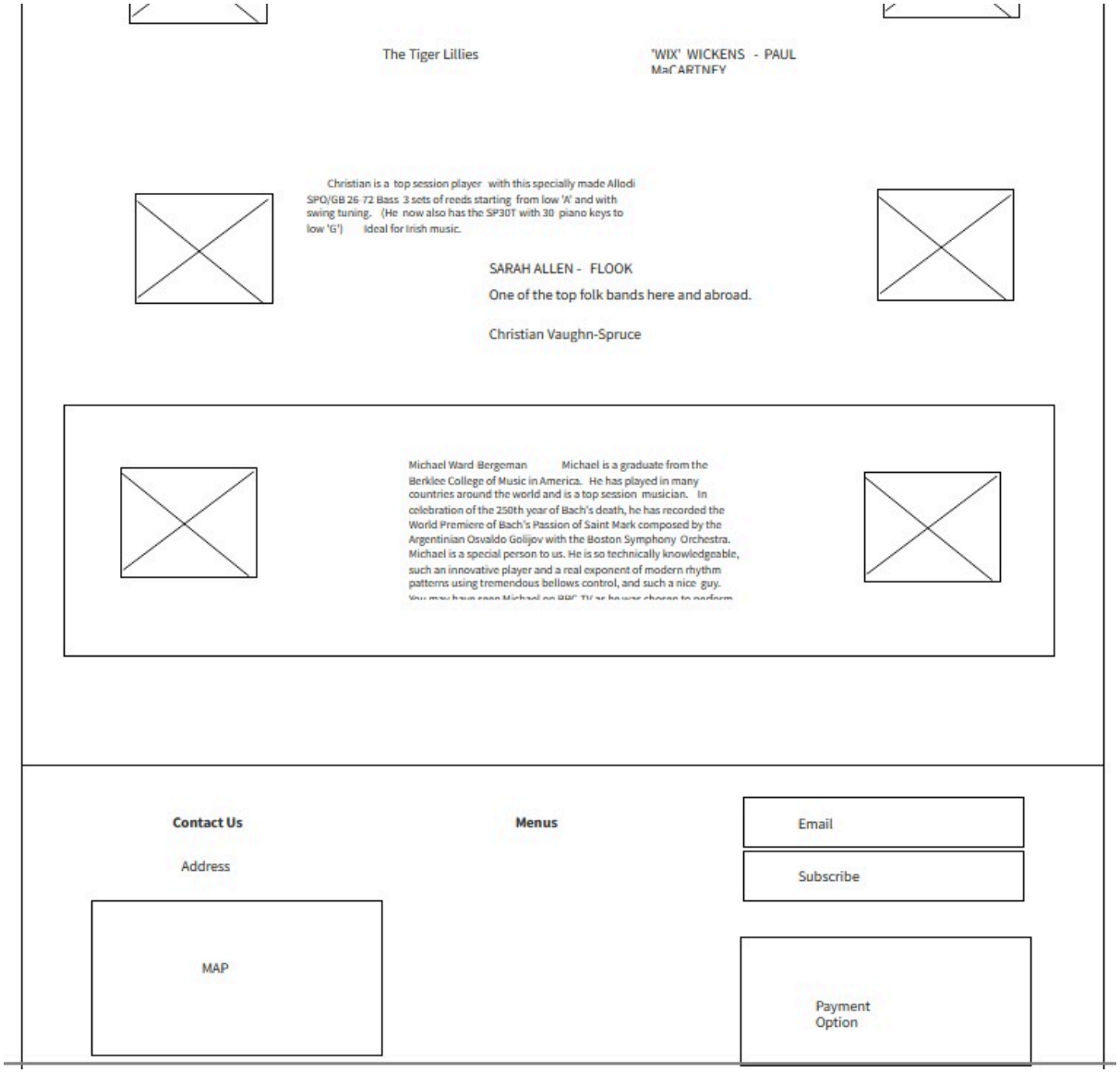
Wireframes

Homepage 1


020 8244 3771
info@accordions.co.uk
Location
Order
USD
Register or Sign in
ALLODI
ACCORDIONS LTD
Search Product
HOME
ABOUT US
Shop
SERVICE
PRICE LIST
TUTOR
FAQ
CONTACT
Favorite
Search product
Category
Category
Category
Category
All Products
Product image
Add to Cart
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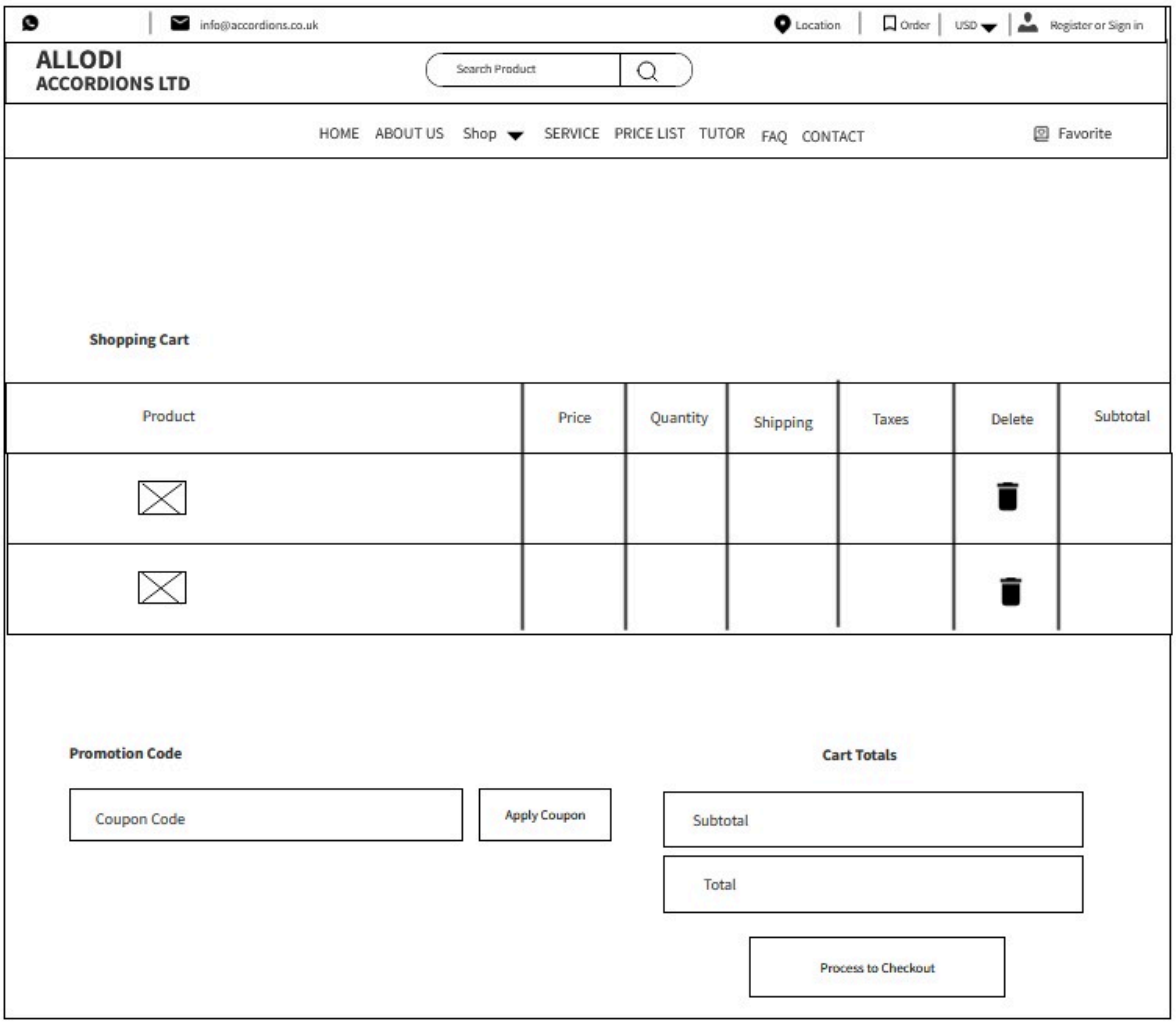

Colors
Product image
Add to Cart
Product image
Add to Cart
Product image
Add to Cart
Product image
Add to Cart
Free Delivery
Button
Price list
The BBC's One Show Featuring Suggs Interview at ALLODI ACCORDIONS LTD
Video
Ideally, we hope you will come to the shop personally, so you may see, and more importantly, try out the different instruments available. It is essential to be able to play, hear and feel the differences between the various models in the same place, so that they may be compared acoustically; instruments can vary quite dramatically in sound depending on a room's size, type of flooring etc.You might like to check out our Allodi/ Fantini player's rogues gallery. The biggest and best advantage of visiting in person of course, is that you will undoubtedly gain from our knowledge and years of experience. However, that said, the website is designed to be the next best thing, allowing you to view our business from the comfort of your own home and in addition, have some vital questions answered that you may have been asking yourself before taking the plunge and purchasing your own accordion. Refer to our section expert advice and Pre War accordions.
Our Popular Products
Get ready to strike a chord with our diverse range of accordions, curated just for you. From classic melodies to modern beats, find your musical soulmate today!
Image
Sold Out
Image
Sold Out
Image
Sold Out


Why We are Best
Free Shipping
Return of
Excahange
Payment
System
Low Price
Top Categories this
Sign up to our Newsletter
...and get your 15% Discount *See Details
Some fine musicians and enthusiasts of the Fantini/Allodi Accordion
This band headed by Martin Jakes is in much demand and tremendously popular around the world. Martin has had three specially made accordions by Fantini, the last one being made to miniature size with five extra bellows folds to allow maximum extension. Pictured left the first custom made Tiger Lillies accordion.

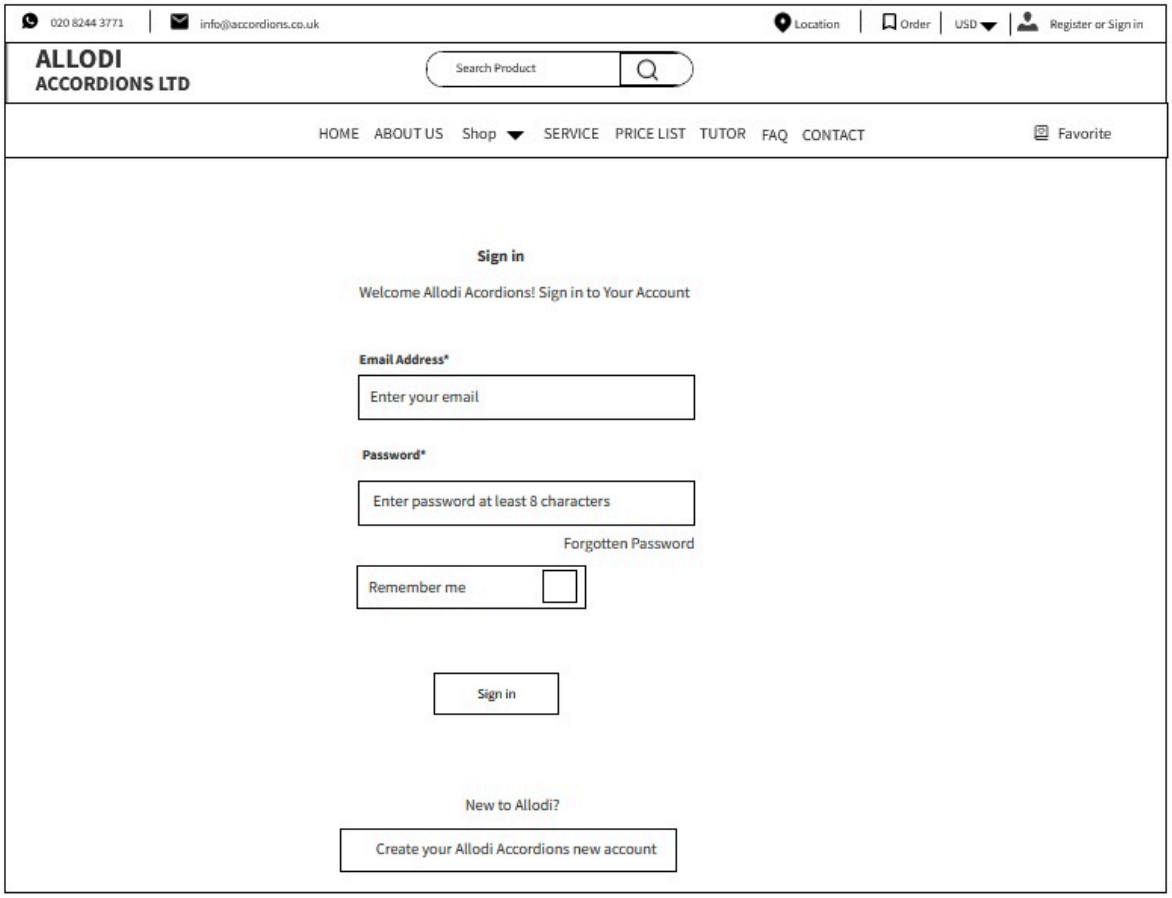

The Tiger Lillies
'WIX' WICKENS - PAUL
Christian is a top session player with this specially made Allodi SPO/GB 26-72 Bass 3 sets of reeds starting from low 'A' and with swing tuning. (He now also has the SP30T with 30 piano keys to low 'G') Ideal for Irish music.
SARAH ALLEN - FLOOK
One of the top folk bands here and abroad.
Christian Vaughn-Spruce
Michael Ward-Bergeman Michael is a graduate from the Berklee College of Music in America. He has played in many countries around the world and is a top session musician. In celebration of the 250th year of Bach's death, he has recorded the World Premiere of Bach's Passion of Saint Mark composed by the Argentinian Osvaldo Golijov with the Boston Symphony Orchestra. Michael is a special person to us. He is so technically knowledgeable, such an innovative player and a real exponent of modern rhythm patterns using tremendous bellows control, and such a nice guy.
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Add to Cart

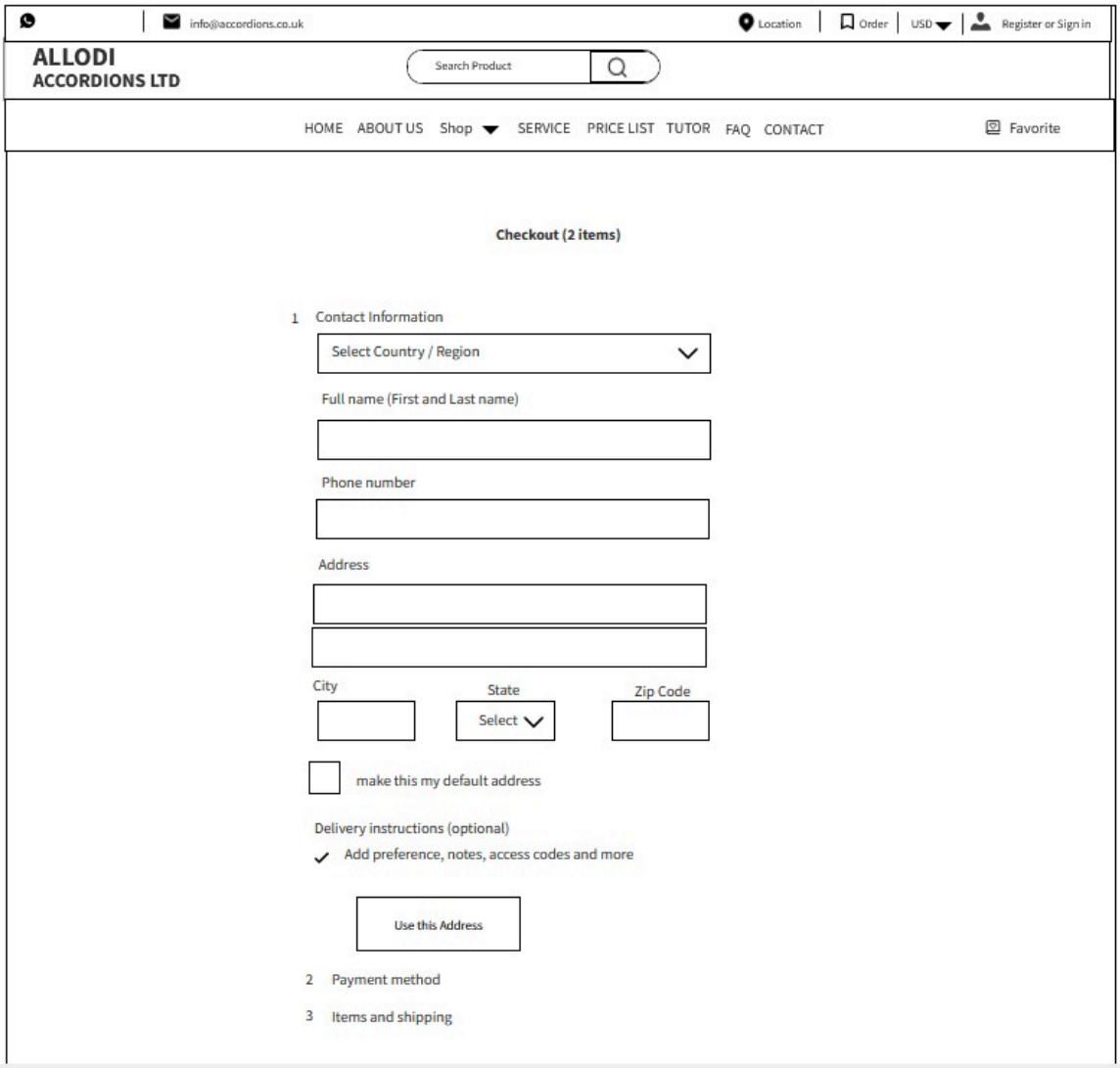

info@accordions.co.uk
Location
Order
USD
Register or Sign in
ALLODI
ACCORDIONS LTD
Search Product
HOME
ABOUT US
Shop
SERVICE
PRICE LIST
TUTOR
FAQ
CONTACT
Favorite
Baroneilli 30 key 48 bass,3 switch Piano Accordion, with Staps, Case, Brown
SKU: ACPK30-BU
Brand : Baronelli
Category: Accordions, Norteno, latin and Tejano , Piano Accordions
1
Add to Cart


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Search Product
HOME
ABOUT US
Shop
SERVICE
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Favorite
Shopping Cart
Product
Price
Quantity
Shipping
Taxes
Delete
Subtotal
Promotion Code
Coupon Code
Apply Coupon
Cart Totals
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Process to Checkout

Sign-in

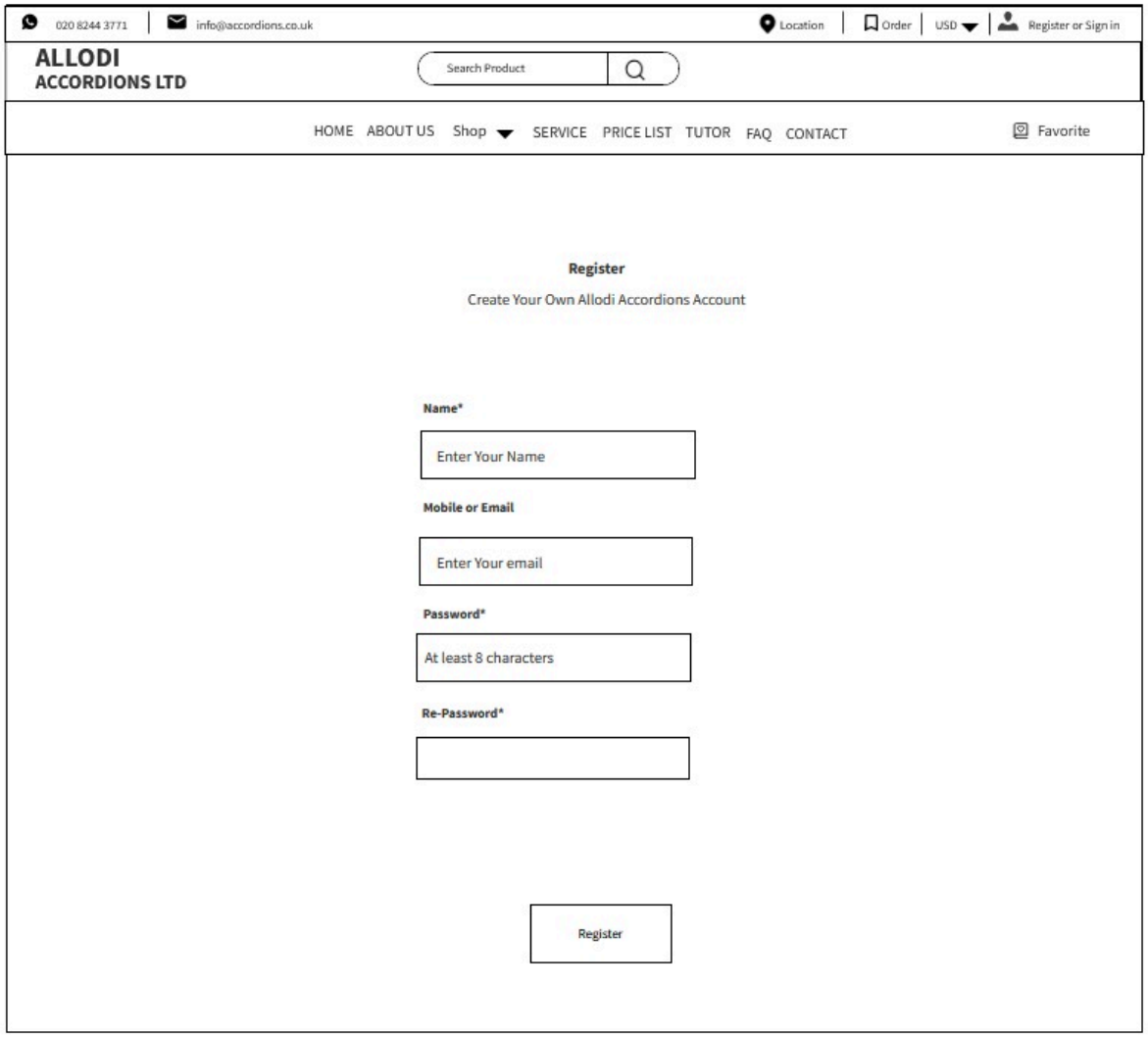

020 8244 3771
info@accordions.co.uk
Location
Order
USD
Register or Sign in
ALLODI
ACCORDIONS LTD
Search Product
HOME
ABOUT US
Shop
SERVICE
PRICE LIST
TUTOR
FAQ
CONTACT
Favorite
Sign in
Welcome Allodi Acordions! Sign in to Your Account
Email Address*
Enter your email
Password*
Enter password at least 8 characters
Forgotten Password
Remember me
Sign in
New to Allodi?
Create your Allodi Accordions new account

Checkout


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Location
Order
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Register or Sign in
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HOME
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Shop
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PRICE LIST
TUTOR
FAQ
CONTACT
Favorite
Checkout (2 items)
1 Contact Information
Select Country / Region
Full name (First and Last name)
Phone number
Address
City
State
Select
Zip Code
make this my default address
Delivery instructions (optional)
Add preference, notes, access codes and more
Use this Address
2 Payment method
3 Items and shipping


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Location
Order
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Register or Sign in
ALLODI
ACCORDIONS LTD
Search Product
HOME
ABOUT US
Shop
SERVICE
PRICE LIST
TUTOR
FAQ
CONTACT
Favorite
Checkout (2 items)
1
Shipping Address
123 KingswayFlat 2BristolBS1 2QT
UNITED KINGDOM
07123 456789
johndoe@example.com
2
Choosing a payment method
Your credit and debit cards
Add credit or debit card> allodi accordions accepts all major cards exxecpt Paypal
Your available balance
Enter Code
Apply
3
Items and shipping

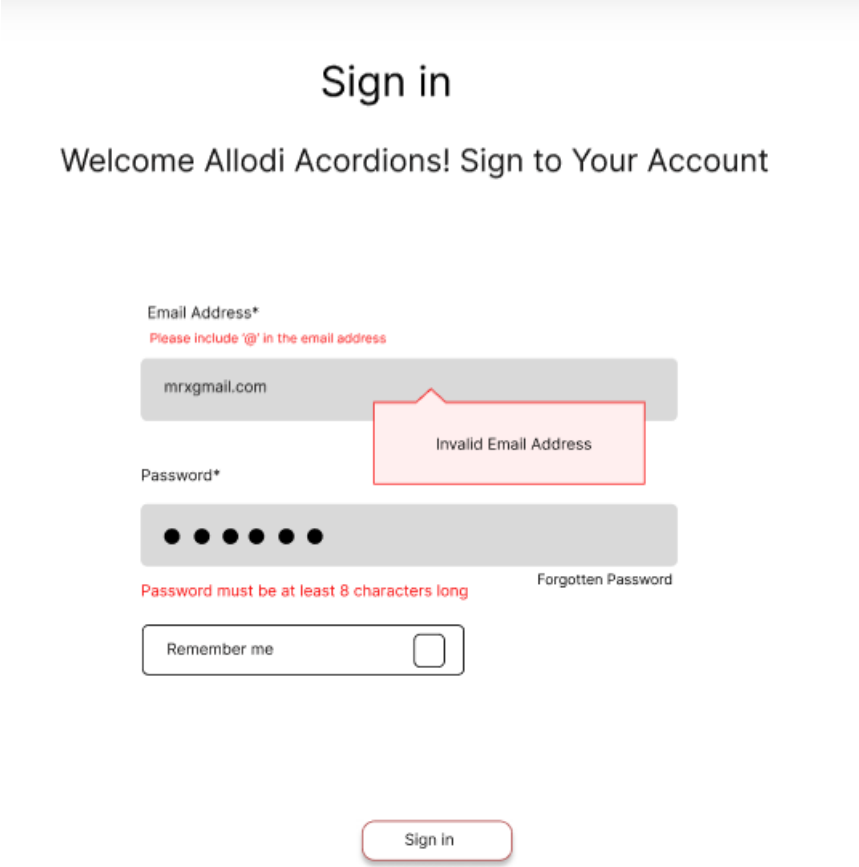
Register



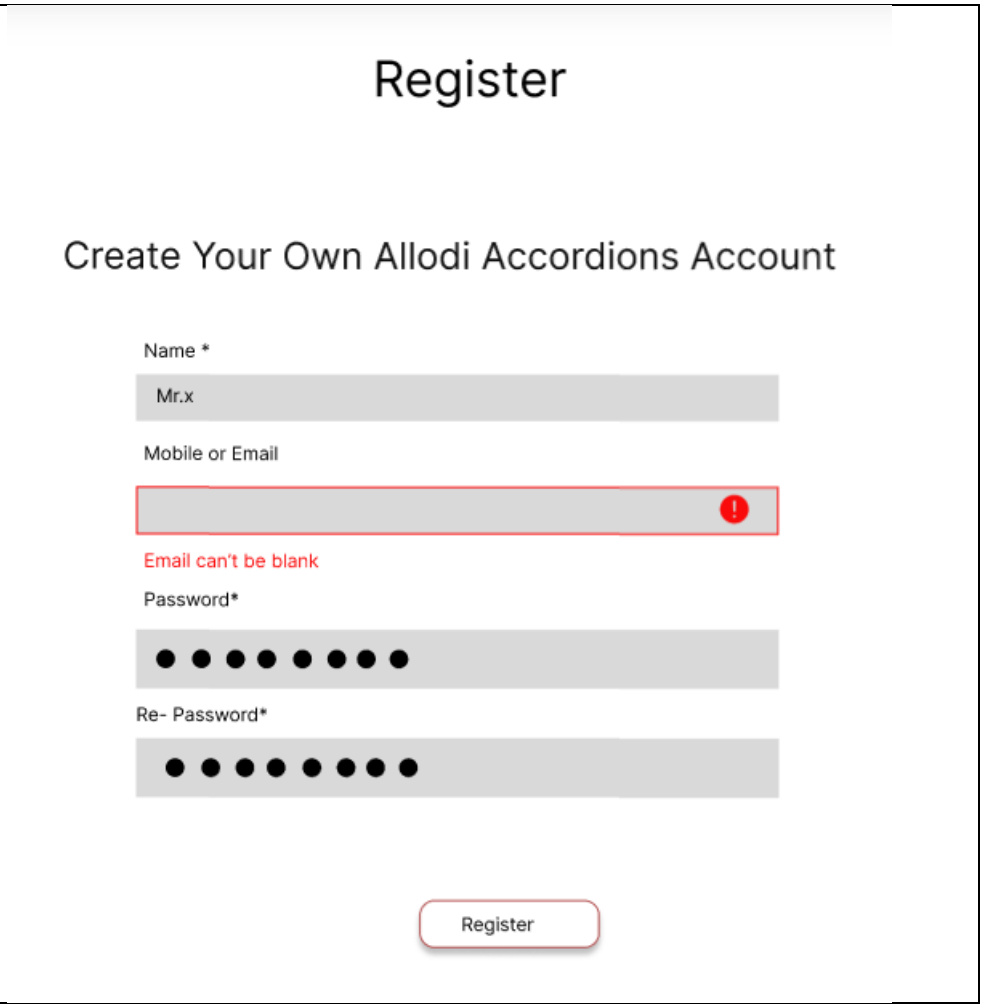
5.4. Testing

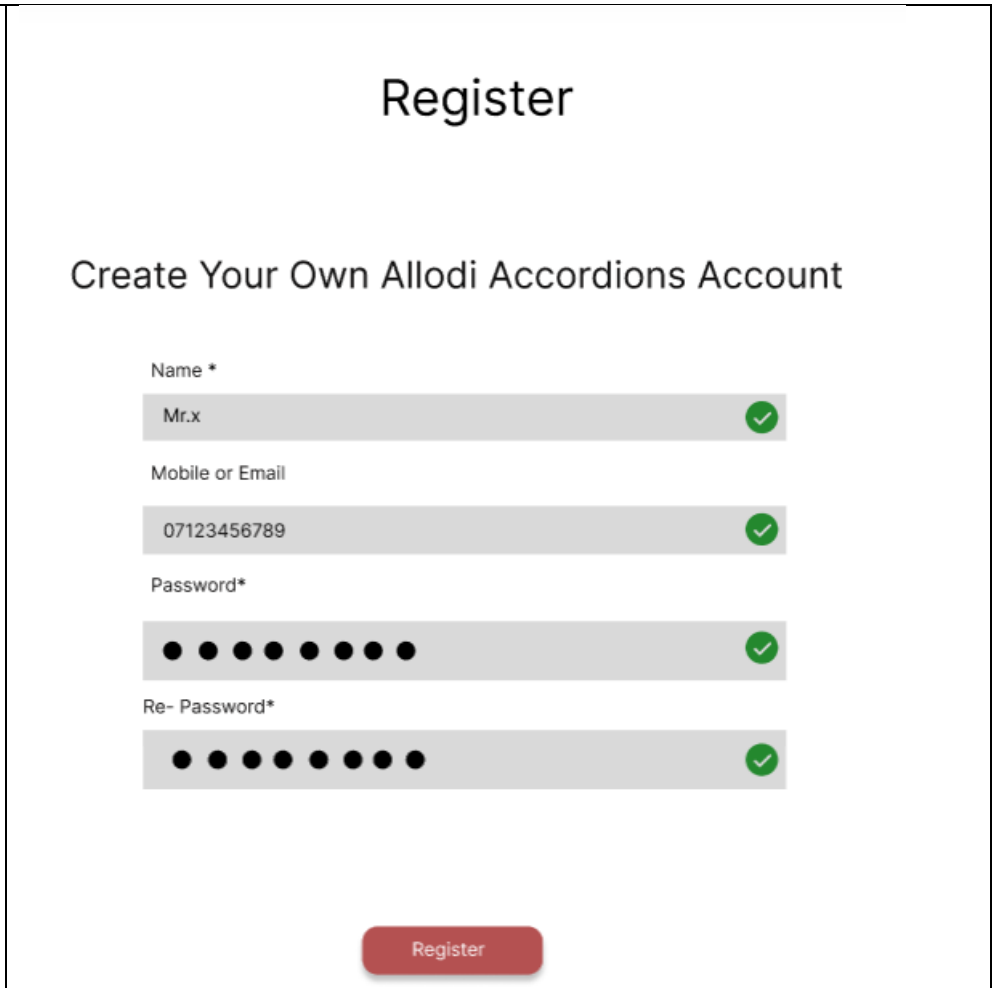
5.4.1. Usability Testing

Test Case ID	1
Description	As a visitor to the web page, a user tries to sign in to the website. However, the user didn't follow the information field criteria according to the web page. The user tries to sign in with an invalid email address and password of less than 8 characters.

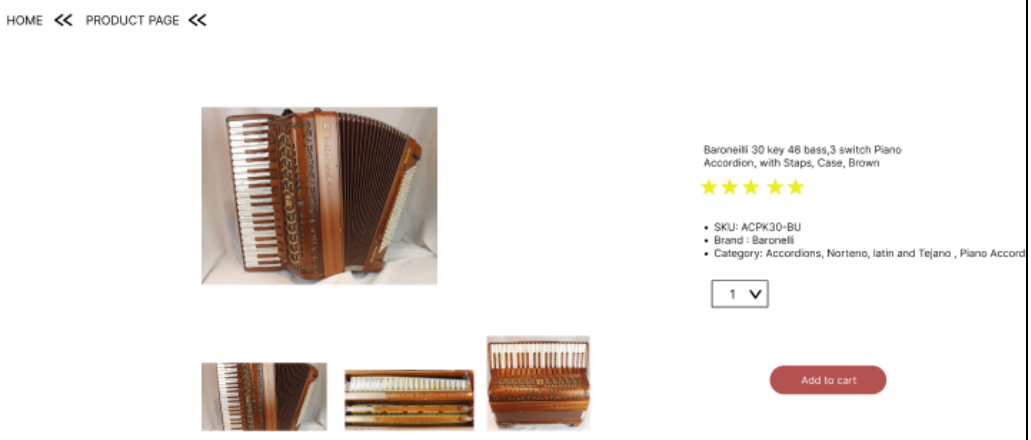
Steps	The user has to visit the sign-in page from the navigation section. The user needs to fill up the information fields with appropriate information. Then the user needs to click on the ‘Sing in’ button
Information field Data	Email Address: Password: Forget password link: Remember password option: ‘Sign in’ button
Expected Result	The ‘Invalid Email Address’ message will show to the user below the email field, and for typing the less-character password the message ‘Password must be at least 8 characters long’ text will be indicated down to the password field.
Interface design	
Result (Pass/Fail)	Pass

Test Case ID	2
Description	As a new visitor to the website, the user wants to register to the web to better access.
Steps	Users can access the registration link from the navigation link. After entering the registration page user has to fill up the information field with personal information. After filling up all the fields with appropriate information the user has to click on the 'Register' button to complete the registration process.
Information field data	Name: Mobile or Email address: Password: Re-password: 'Register' button
Expected result	After filling out all the fields with appropriate information 'Register' button will be activated automatically and the user will complete the registration process successfully. If any of the information fields have been filled with inappropriate info then the error message will appear below the field with a guided message.

Interface Design	

	
Result (Pass/Fail)	Pass

Test Case ID	3
Description	The user wants to view a product from the product page.
Steps	Users can enter the product page from the navigation link. To enter the product page user has to click on the 'Shop' link from the navigation area. After clicking on the 'shop' user will be landed on the product page. There users will be able to see all the products listed. To view a product user needs to click on the area of that specific product, after the click system will take the user to that specific product page.

Information field data	Picture of the product Details text of product description Review's using the star icons Quantity selection option 'Add to cart' button
Expected result	• Users will be able to see the details of the product on this page. • The user would be able to shop the product from this page. • Users can return to the previous page by using the 'product page' link. • User can return to the 'Home page' from this page
Interface Design	
Result (Pass/Fail)	Pass

Chapter 6. Evaluation

6.1. Evaluation of the designed system using Nielson's Ten Heuristics

The website design system that was built for this research followed Nielsons ten heuristics.

Visibility of System Status

Nielsen's first principle is to inform users about their actions. If the user gets informed about the status of their action and their past interactions then it will determine users about what their following stages will be. When users find a design predictable it builds their trust in that product (Lebson, 2021).

In the re-design of the Allodi Accordions Ltd website, there's has been used some of the new elements keeping in mind user requirements.

Cart Option

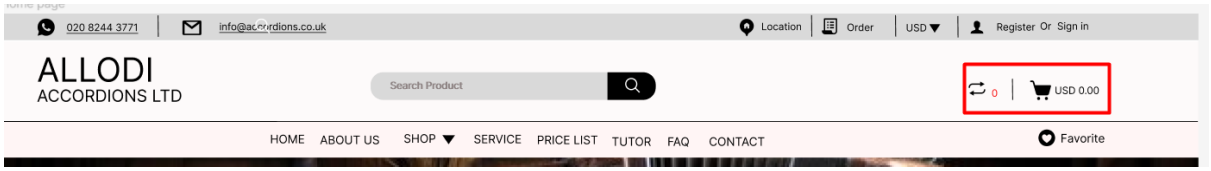


Figure 33: Cart option on Allodi Accordion Ltd website (Source, self-made).

The cart option on the website increases customer satisfaction, it will inform the user about their purchase status and lets them easily add and remove products from the purchase list.

System Errors

On the re-design of the interface, the user will be informed about the system errors. If the user types the wrong info, the system will inform the user about the errors and will acknowledge the user with guidelines.

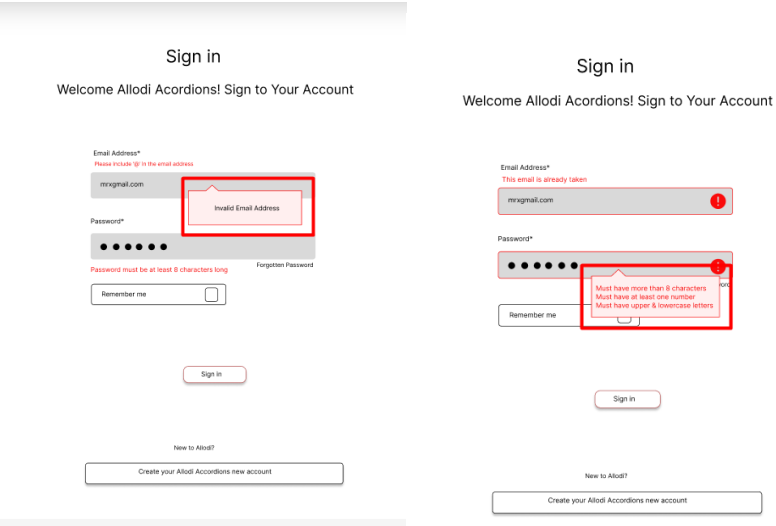


Figure 34: Interface Showing users with system errors (Source, Screenshot).

Loading Indicators

Loading indicators were added in the interface design as the element informs and engages the user during the waiting period.

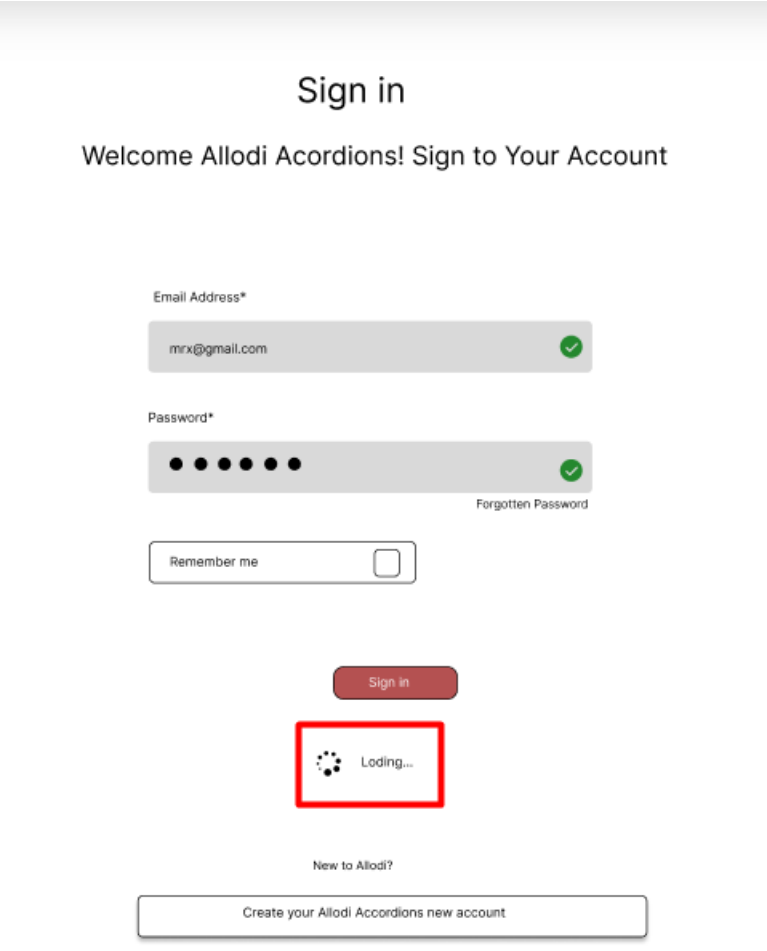


Figure 35: Usage of loading indicator on the website re-design (Source, Screenshot).

Match between the system and the real world

The second heuristics of Nielsen explains that the system should state the user's language and follow real-world conventions. For making a connection with the real world the components must look in reasonable instruction so that it makes sense to the users according to their life knowledge. Keeping in mind the elements that were added to the website re-design like

Navigation link

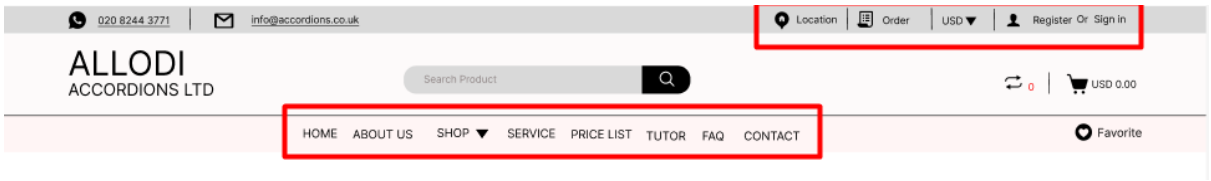


Figure 36: Navigation bar menu from website re-design (Source, Screenshot).

The features that were added to the navigation menu re-design reflect the user's mental model and terminology that they are familiar with on the e-commerce website. The purpose of the navigation menu is to ensure user finds their need without confusion.

Search Engine

The search engine is acknowledged by everyone, it helps users find specific needs within a short time and it builds great user experience.

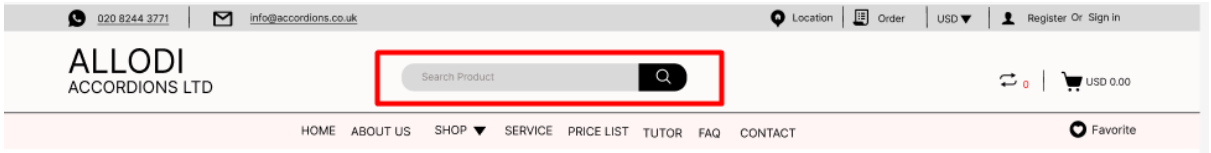


Figure 37: using the search engine on re-design (Source, Screenshot).

Form Fields:

In input forms of re-designing, some labels and placeholders define the information required clearly and familiarly. This benefits users in recognising what is expected of them and decreases errors during data entry.



Figure 38: Input forms of re-designing (Source, Screenshot).

User Control and Freedom

In the interface design users should get the freedom to decide and take actions according to their needs. The system should not make decisions for the users, it only can guide them with the path that they can take. The interface design should have the option for users by which user can undo or redo their actions.

Clear Exit Paths

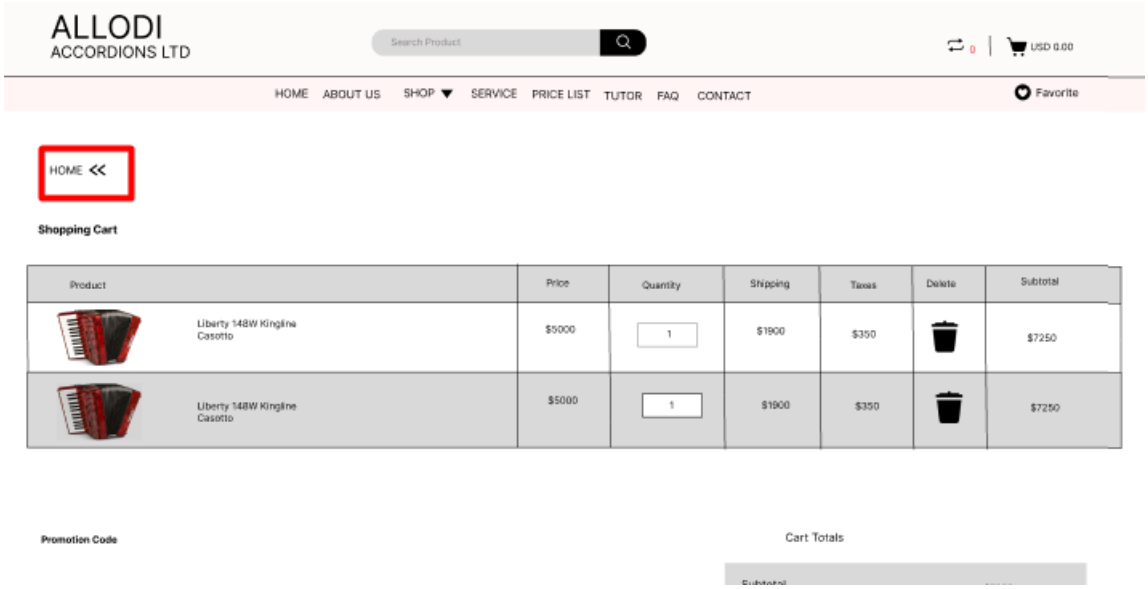


Figure 39: Usage of exit path on the website re-design (Source, Screenshot).

When users initiate a multi-step process such as filling out a form or making a purchase, the re-design adds a projecting "Home" button or link that will allow the users to abandon the procedure and return to the previous page like “back to cart”.

Confirmation Dialogue

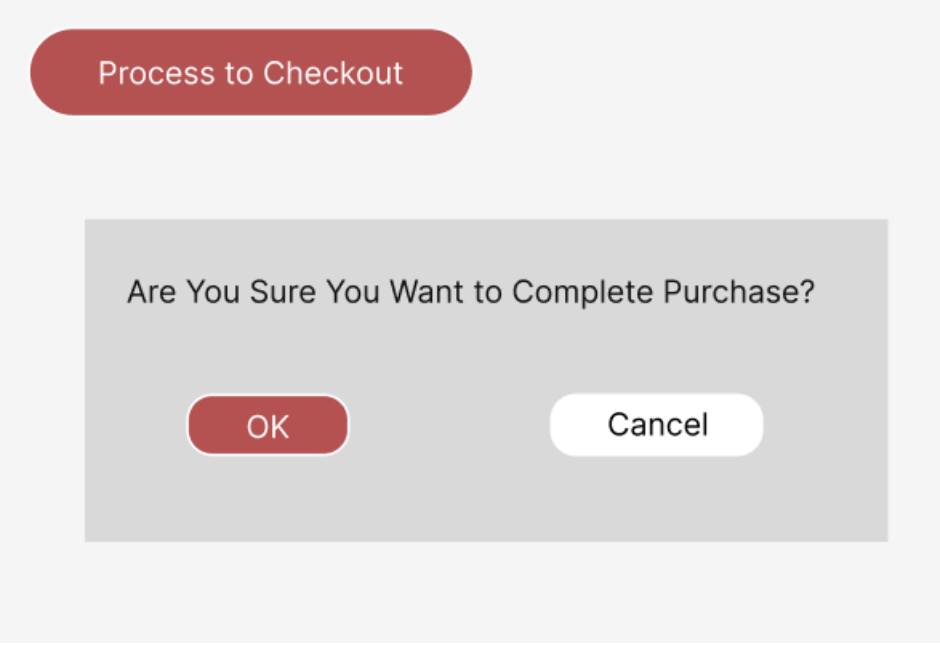


Figure 40: Confirmation dialog showing on the interface (Source, self-made).

Confirmation dialogue helps users to be reassured about their actions. The actions are like deleting data, asking for confirmations and logging out from the portal. Confirmation dialogues allow users a chance to reconsider their actions before proceeding further.

Consistency and Standards

It creates a good user experience if the user gets to see the same language throughout the whole system. If a user gets consistent language throughout the design, it will avoid user confusion.

Navigation Menu

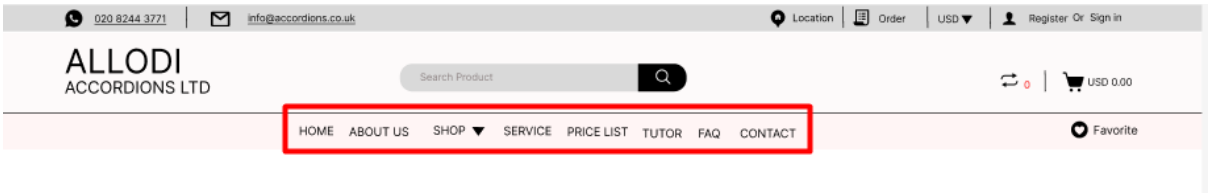


Figure 41: Navigation menu (Source, Screenshot).

The redesign of the navigation menu structure will be consistent across all pages of the website. A horizontal navigation menu at the top of the website with categories like "Home," "About Us," "Shop," and "Price List," maintains this same structure and placement on every page to provide users with a predictable and intuitive navigation experience.

Error Prevention

Error messages are important for letting users know about the limitations of the system. In a system, there are two types of errors Slips and mistakes. Slip errors are caused by inattention and mistakes are conscious errors.

Clear Error Messages

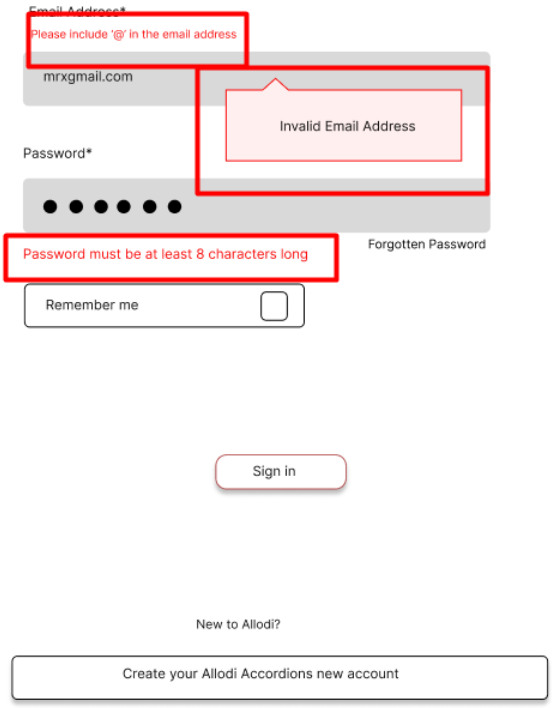


Figure 42: Interface design showing error message to user (Source, Self-made).

If a user attempts to submit the form with invalid inputs, system will provide clear and specific error messages that highlight the exact fields with errors and explain how to correct them.

Recognition Rather than Recall

Foe users it is difficult to keep the memory, a good interface design lessens the user's memory weight by making elements, actions, and options noticeable. Information is compulsory to in the design that should be observable or easily retrievable in the time of need.

Dropdown Menu

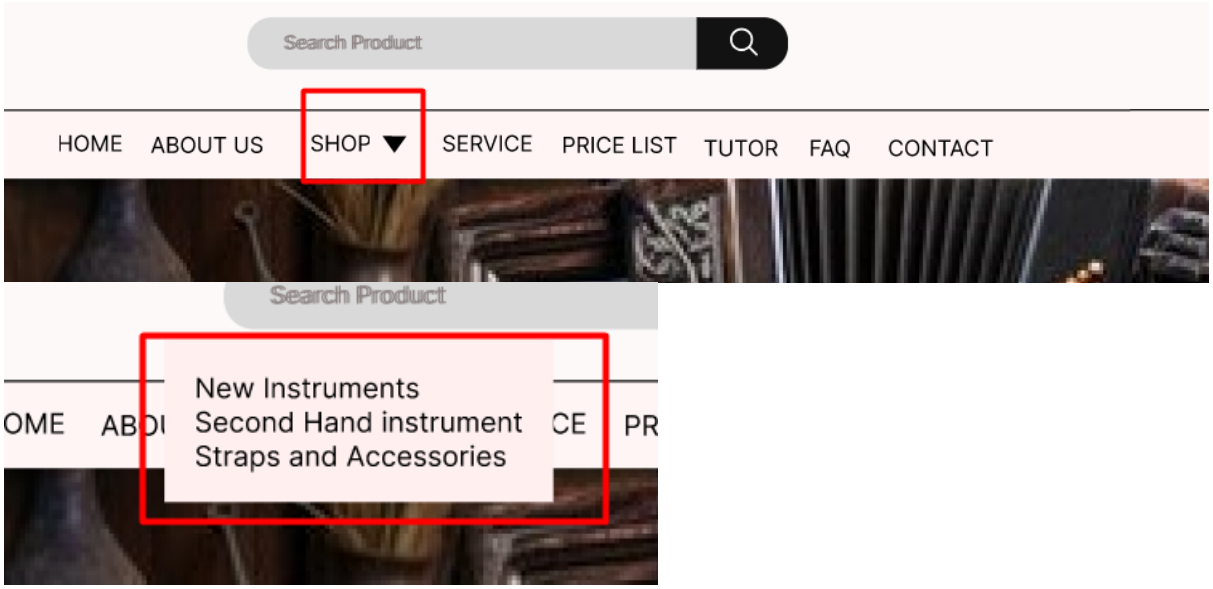


Figure 43: Dropdown menu on the navigation link (Source, self-made).

Within the "Shop" section, use dropdown menus or expandable panels to organize related options and features. For example, the "Shop" dropdown menu can include options for New Instruments, Second Instruments, Straps & Accessories.

Clear Labels and Icons

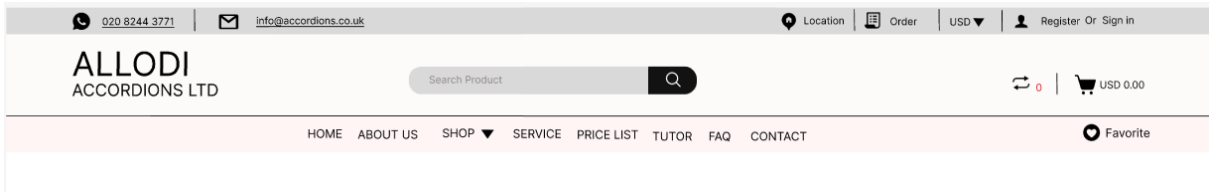


Figure 44: Clear labels and icons on the navigation link (Source, self-made).

In the re-design of the website, descriptive labels and icons is used to represent different account management functions.

Recall in User Interface

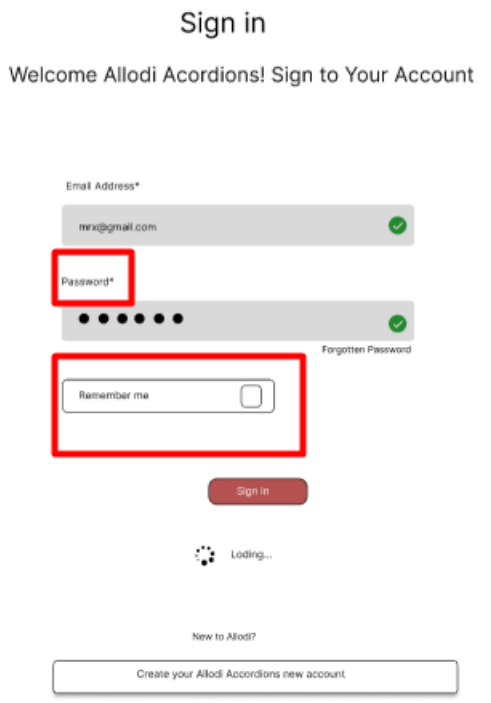


Figure 45: Recall in user interface (Source, self-made).

The classic example of recall in an interface is sign-in. When you sign in to a site, you must remember both a username (or email) and a password.

Flexibility and Efficiency of Use

For speeding up both inexperienced and experienced user interaction using elements like 'Shortcuts' will allow users to tailor frequent actions. Flexible processes can be carried out in multiple ways so that users can decide whichever method works for them.

Website shortcut Button

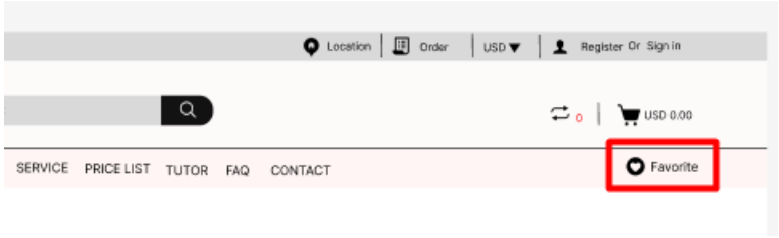


Figure 46: Using the shortcut button on the interface (Source, self-made).

Users can see all their searching products and favourite product by clicking on the 'favourite button'

Aesthetic and Minimalist Design

On an interface, the design elements should keep the contents focusing on the element that is essential for the visual design. It is necessary to ensure that the visual elements of the interface support the user's primary goals.

Visual Hierarchy

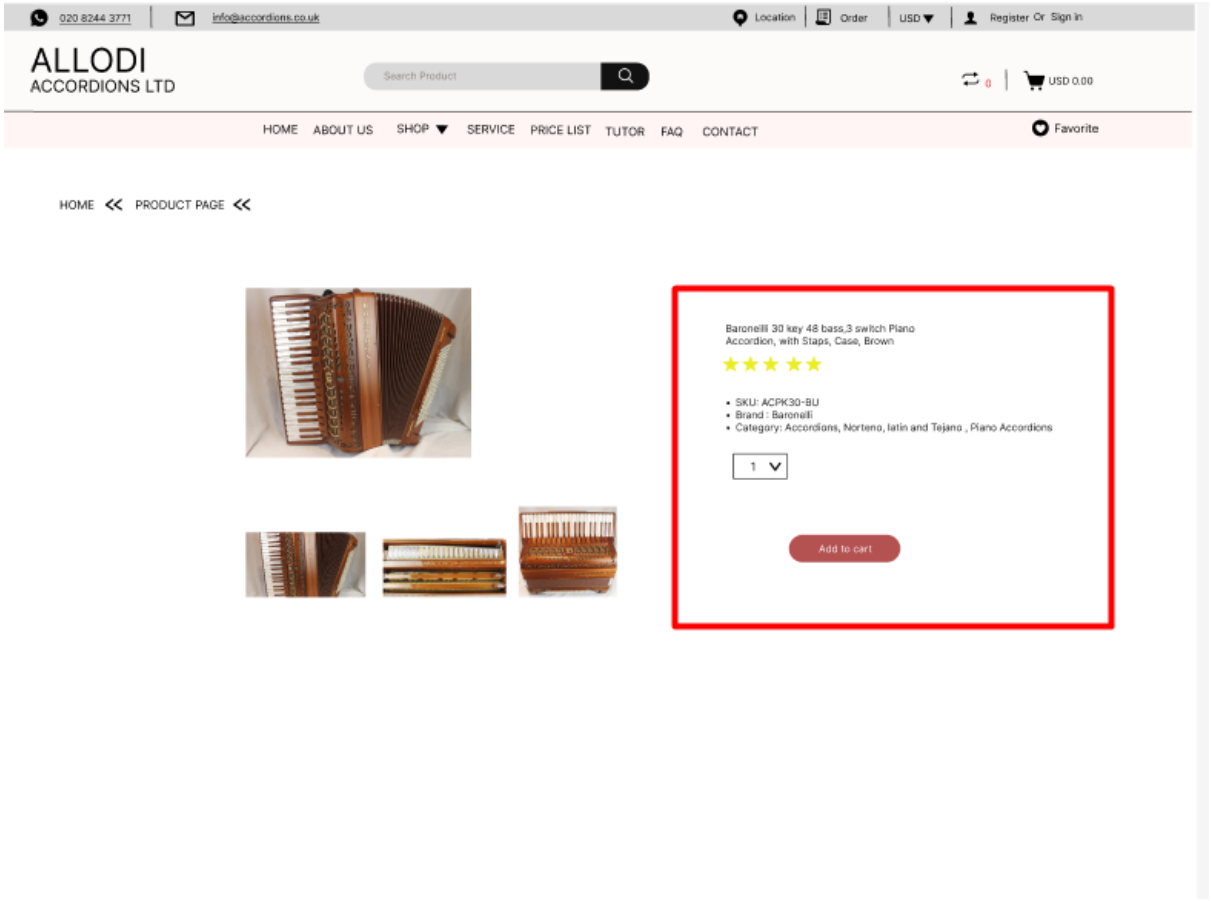


Figure 47: Visual Hierarchy on the interface (Source, self-made).

On the re-design of the website of Allodi Accordions Ltd, it used clean and organised layouts that prioritize important information like product names, prices, and pictures. The design also followed consistent typography, colour schemes, and spacing to create a visually pleasant design.

Using More breathing space in the design

It is proven that using white spaces in interfaces is a great user experience. It made the design very spacious and easy to focus users on the Important topic.

Help Users Recognize, Diagnose, and Recover from Errors

Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.

Real-time Validation



Figure 48: interface showing real-time validation (Source, self-made).

If the user enters an incomplete or invalid email address (e.g., missing email address), the form field is highlighted in red, and an error message appears next to it indicating the issue.

Clear Error Message

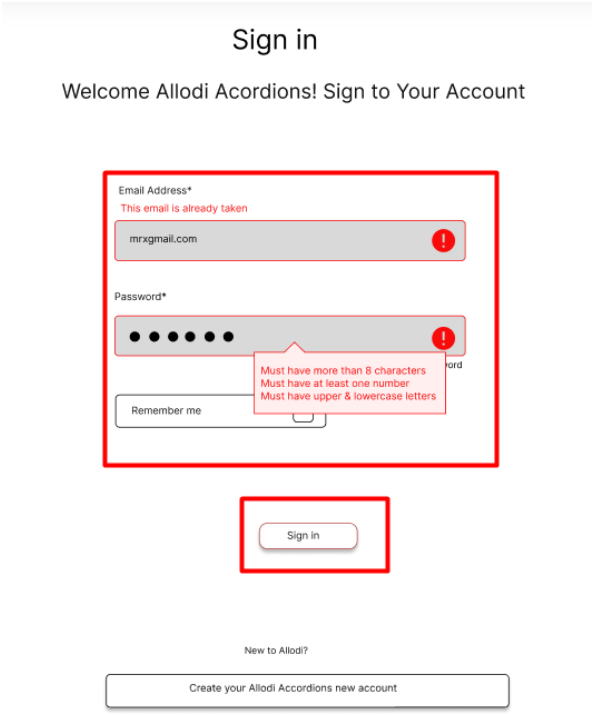


Figure 49: interface showing clear error messages (Source, self-made).

When the time of user puts a wrong password, the system will show the user the error and help them with the guideline.

Help and Documentation

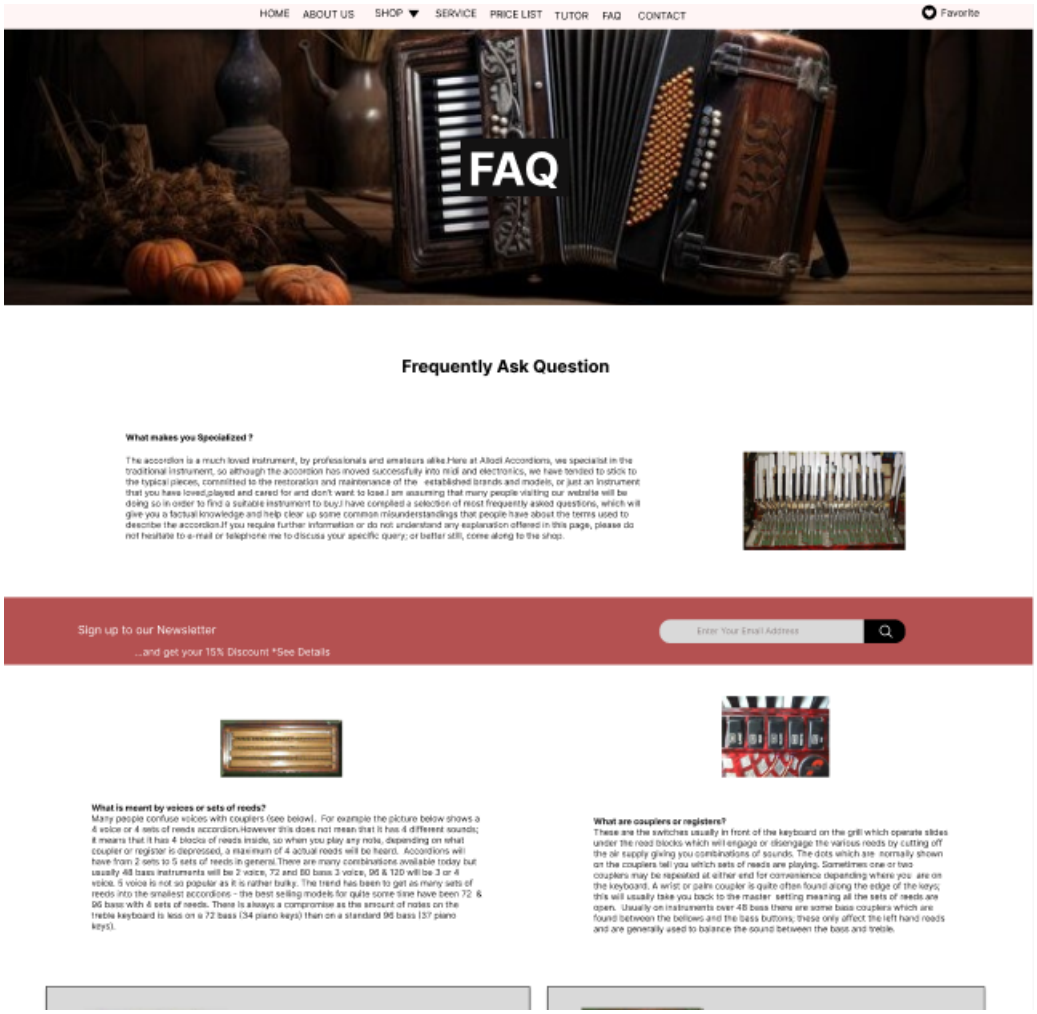


Figure 50: Help and documentation page on the interface (Source, self-made).

It is most excellent if the constitution doesn't necessitate any supplementary explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

6.2. Evaluation of the designed system using WCAG 2.2

The WCAG documents explain how to make web content more accessible to persons with susceptibilities. The idea of WCAG is that web content should be comprised with recognised information such as text, images, sounds, code or rise that defines structure, presentation, etc (Miller, 2021). For the secondary evolution of the website design here, the research has followed the WCAG conformance pointer level of A and AA.

Perceivable

All the information that is used on the interface as components has been presentable in ways that users can perceive.

1.1.1 Non-text content (A)

Each input form section including the buttons has a name that describes its purpose.

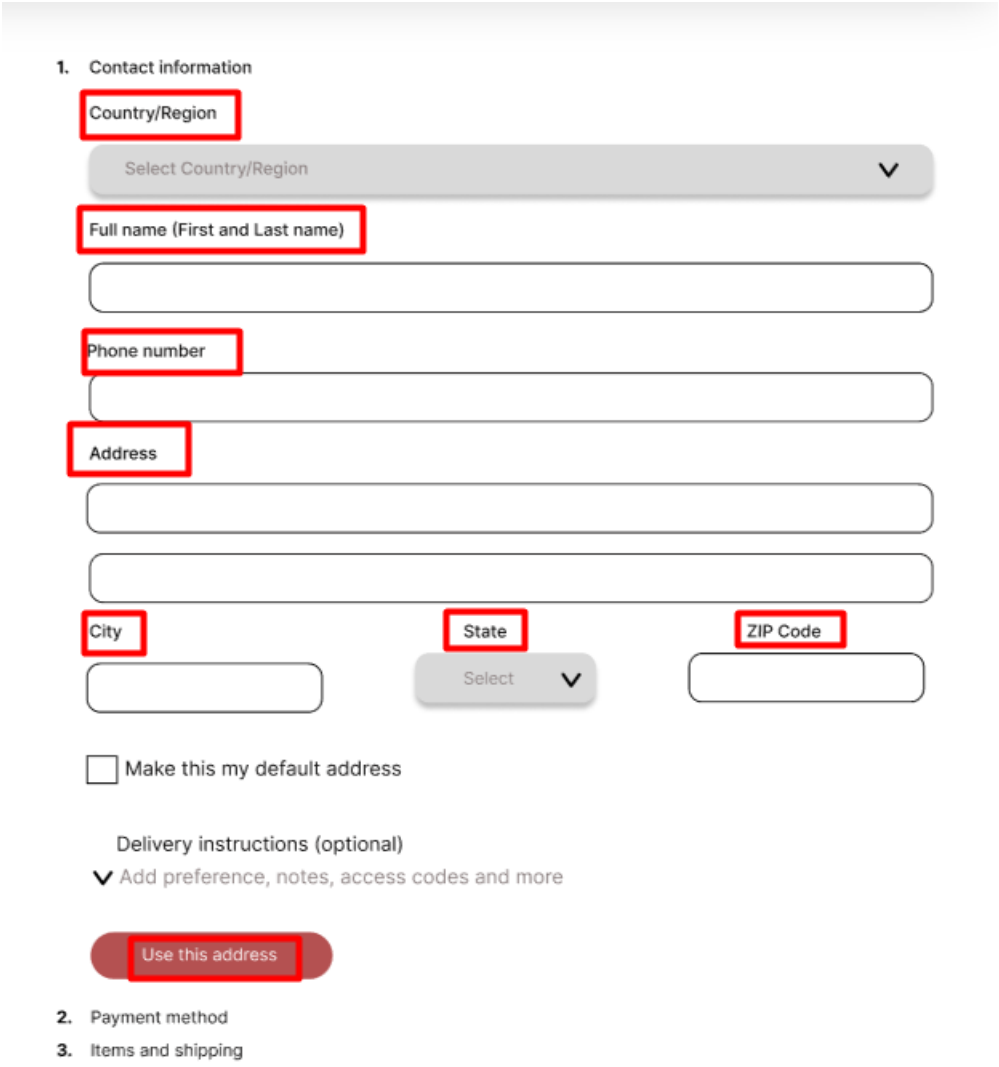


Figure 51: Reference picture of non-text contents. (Source, self-made).

1.4.1 Use of colour (A)

Colour has not been used as only visual information conveying or indicating an action of the website.

1.4.3 Minimum contrast (AA)

The text and the background have a minimum ratio of 7:1.

1.4.12 Text Spacing (AA)

The line height is 1.5 times the font size.

Operable

All the components that are used on the interface's navigation link are accessible. The principle of navigation has been followed by including an informative title on every page.

2.4.3 Logical Focus Order (A)

The features of each page like buttons, links, Information fields, etc. are used in a logical order. For example, In the sign-in process while the user fills in the information field the sign-in button will show as an inactive button unless the user fills the information field with the correct info. After getting the correct info the sign-in button will be active automatically.

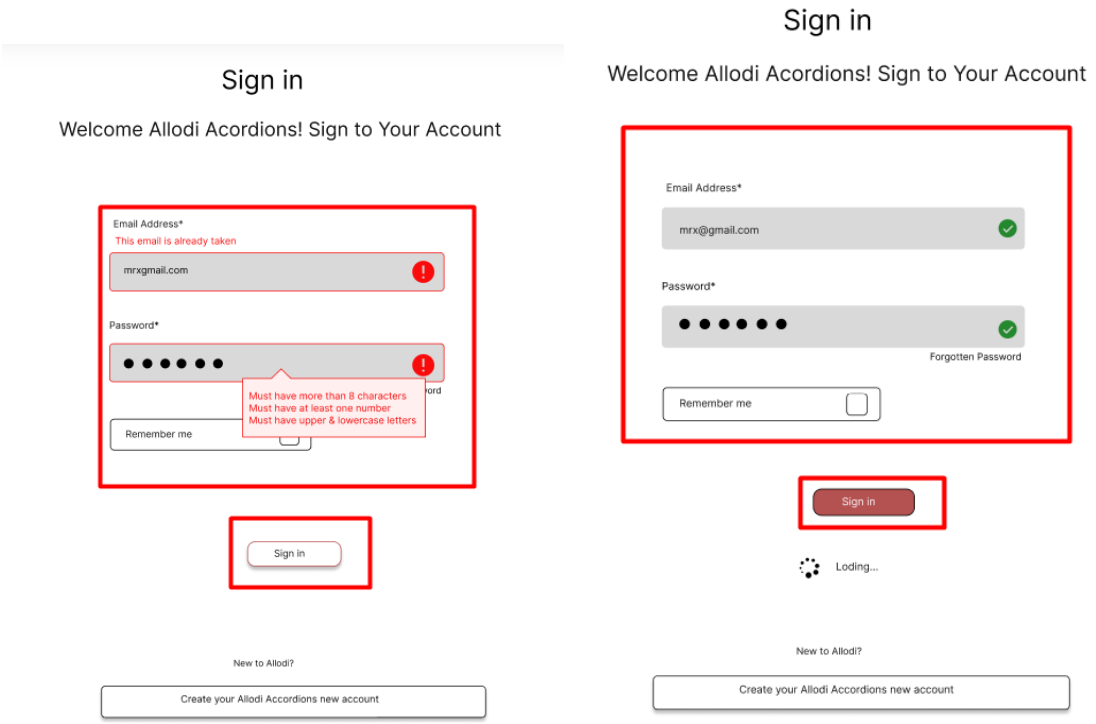


Figure 52: Reference for logical focus order (Source, Self-made).

2.4.6 Headings and Labels (AA)

The heading section of each page clearly shows the purpose of the contents.



Figure 53: the heading shows the clear purpose of each web page (Source, self-made).

2.4.11 Focus Not Obscured (Minimum) (AA)

An easy way is provided for the user to return to the Product page without losing their place.

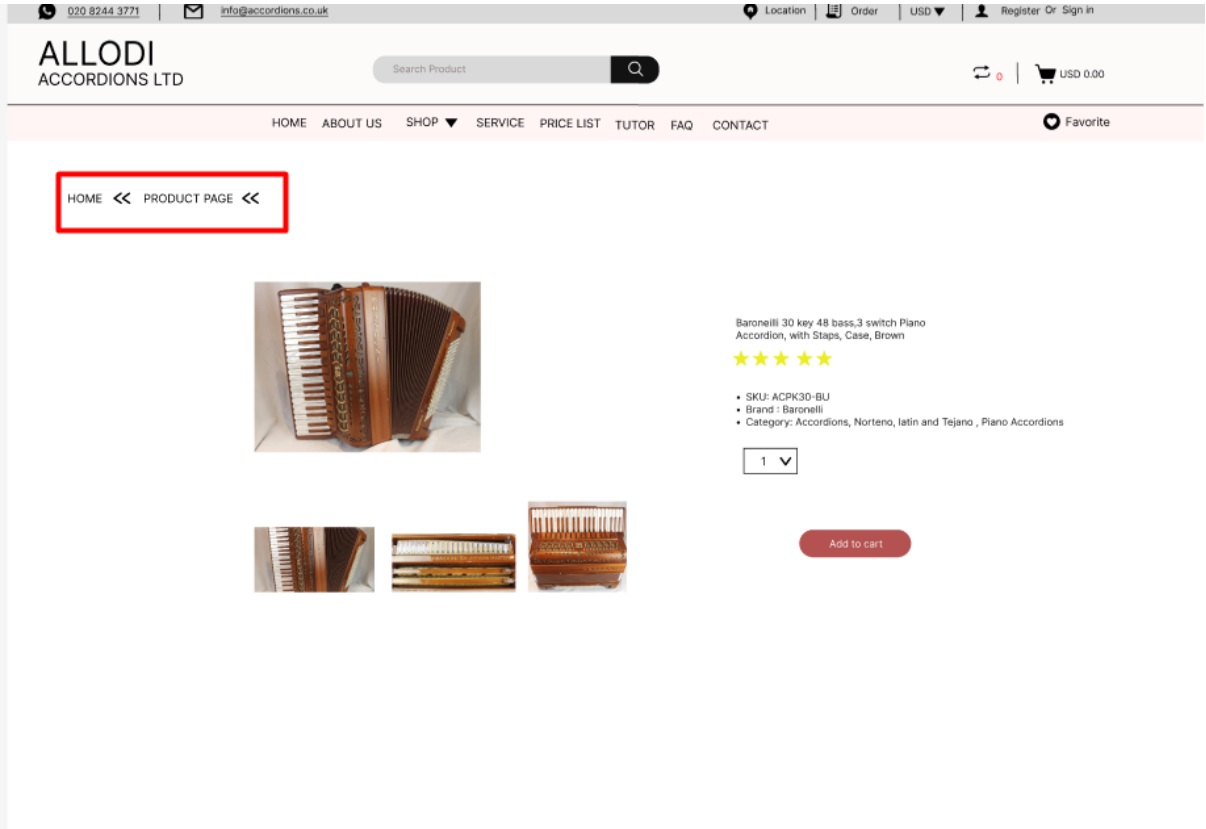


Figure 54: A clear way for the user to return to the previous page (Source, Self-made).

2.5.1 Pointer Gestures (A)

All multipoint gestures for operation can be operated with a single pointer.

Understandable

The interface design is done in a way that is easier for the users to understand. All the text content that is used on the website's interface re-design is readable for the users.

3.1.1 Language of Page (A)

All the web page text contents are human-readable.

3.1.2 Language of Parts (AA)

The text of each passage in the content area is human-readable

3.2.1 On Focus (A)

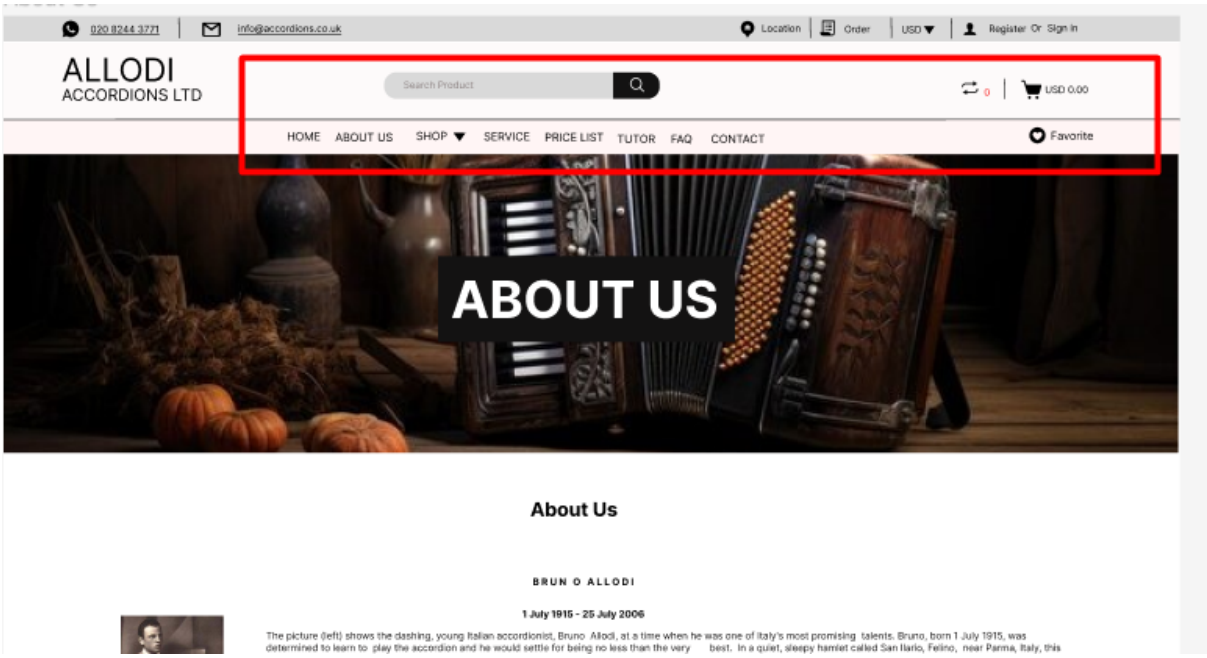
When the user chooses any component, it doesn't initiate a change in the whole context.

3.2.2 On Input (A)

If the user selects any of the input sections it does not change automatically unless the user has been advised before using the component.

3.2.3 Consistent Navigation (AA)

Throughout the home page, the navigation link would be consistent for easy access to the important features for the users, unless the user initiates a change.



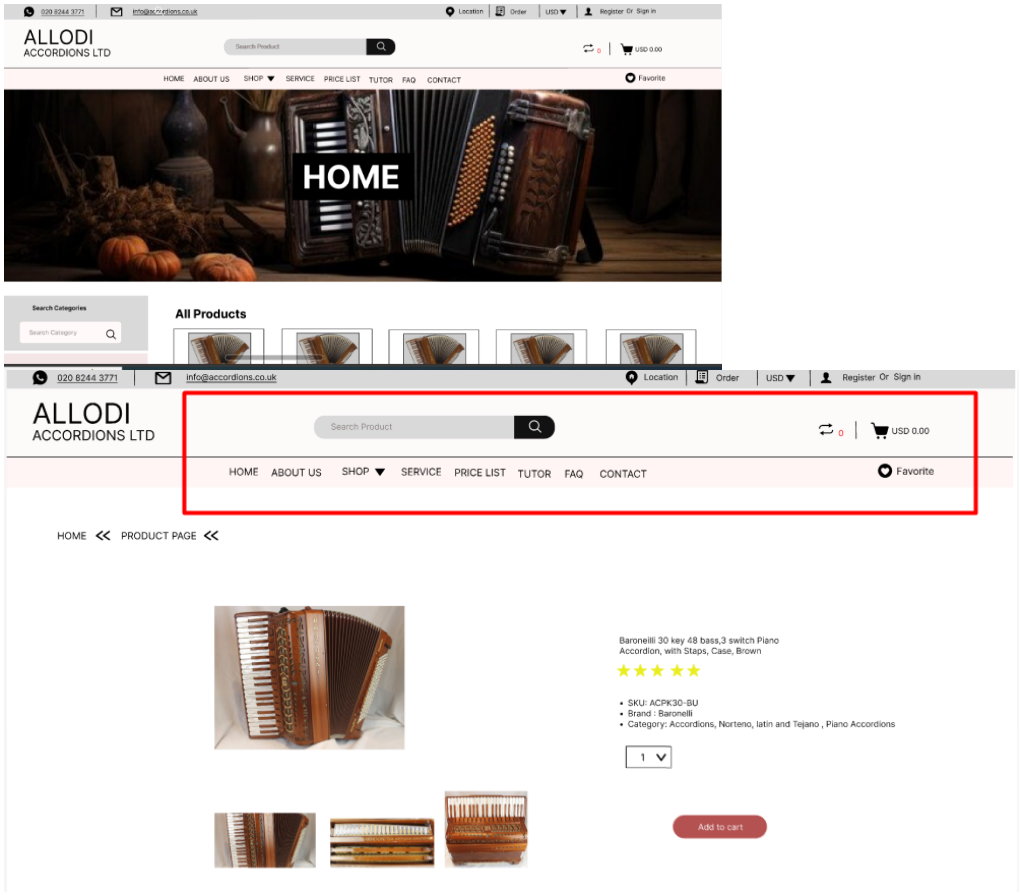


Figure 55: Consistency of the Navigation link throughout different pages. (Source, self-made).

3.2.4 Consistent Identification (AA)

Components with the same functionality within a few web pages will be identified as the same components with each click consistently.

3.3.3 Error Suggestion (AA)

User's information entry will be checked and if required necessary suggestions will be provided.

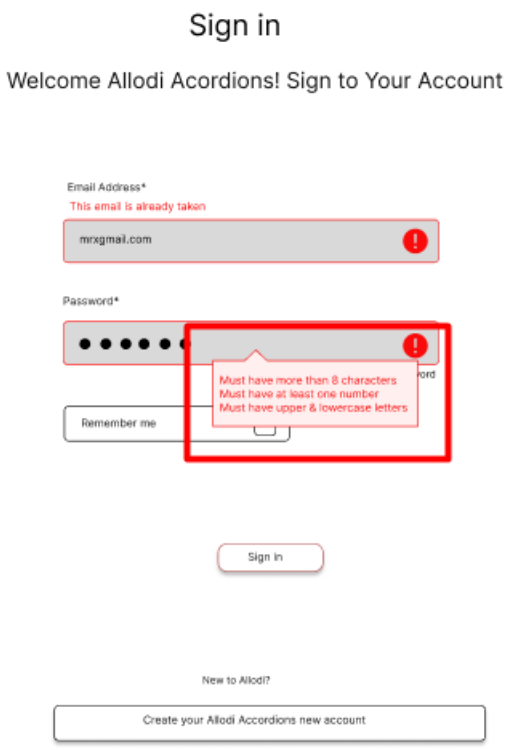


Figure 56: System showing error suggestions (Source, Self-made).

Robust

A wide variety of users can interpret all the content.

4.1.2 Name, Role, Value (A)

The components like buttons, links, etc. of the user interface are programmatically determinable by users like keyboard-only users, screen-reader users, etc

4.1.3 Status Messages (AA)

For the content implementation, it used markup languages without receiving user focus.

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